Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

Author names to be supplied for arXiv v1

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Abstract

Latent predictive world models such as JEPAs avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We introduce action-conditioned predictive consistency (ACPC) as a no-retraining diagnostic for JEPA-style control models: paired clean/corrupted histories of the same state must induce consistent action-conditioned rollouts and candidate costs, while a local task-feature proxy margin checks that action-relevant differences remain separated. The analysis gives three calibration links for this diagnostic chain: fixed-candidate cost-drift and top-1 stability, a sampled-pool bound separating ACPC-tail and clean-margin terms, and a local Gaussian expansion showing that the measured basin radius is action-conditioned encoder–predictor sensitivity. On PushT, TwoRoom, Reacher, and Cube, no-noise LeWM is fragile under observation-only Gaussian noise (PushT 86.33% → 4.33%, Reacher 58.67% → 18.33%), while full-sequence input-side noise training supplies recovered checkpoints across training seeds 3072/3073/3074. The diagnostic evidence then asks whether that recovery is selective: a seed-3072-fixed ACPC-family rule localizes broad held-out plateaus (7/8 held-out blocks within 5 points) without dominating simple plateau baselines; a margin-conditioned fixed-pool audit reduces top-quartile noisy top-1 flips to 0.00 at the fixed high-noise endpoint; and local proxy margin pass-rates rise from 0.53–0.81 to approximately 1.00. Together, these results define an auditable matched-Gaussian diagnostic study for fixed JEPA checkpoints, with PLDM replication, negative target-view and heteroscedastic ablations, and blur/resize checks used to bound scope and failure modes.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: `https://github.com/Anguo-star/lewm-acpc-diagnostics`

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [2, 4, 39]. For control, however, "irrelevant" is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model's *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. A no-noise LeWM [26] checkpoint scores 86.33% on unperturbed PushT but only 4.33% under observation-only Gaussian noise at std 0.08 with the goal unperturbed; Reacher drops from 58.67% to 18.33%, and TwoRoom from 94.00% to 65.67%. Observation+goal noise is reported as a harder auxiliary stress condition. Test-time visual fragility is not new [21, 43, 16, 28]; our focus is what it reveals about JEPA latent world-model control.

1.2 Operationalizing action-conditioned predictive consistency

We operationalize a downstream-control view of visual robustness for JEPA world models through *action-conditioned predictive dynamics.* Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a perturbed history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{t:t+k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k\big(z, \mathbf{a}_{t:t+k-1}\big) \;\approx\; F^k\big(\tilde{z}, \mathbf{a}_{t:t+k-1}\big), \qquad k = 1, \ldots, H, \tag{1}$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This diagnostic view changes what counts as a useful robustness measurement. Encoder geometry is a useful first-stage diagnostic because it shows whether visual perturbations enter and reshape the latent neighborhood, but encoder-level latent closeness, $\|z - \tilde{z}\| \to 0$, is not a complete control-robustness definition: large nuisance shifts can wash out after prediction, whereas small shifts can still be amplified downstream. Planning, cost, and action metrics remain useful as *downstream readouts* of predictive consistency; future objectives should regularize action-conditioned predictive dynamics while preserving action-relevant discriminability.

1.3 Existing noisy-evaluation results as a controlled probe

We use input-side noise augmentation as the controlled intervention. The $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$ sweep across four tasks creates a reproducible grid of paired baseline/recovered checkpoints for asking where closed-loop recovery appears in the encoder–predictor–planner chain. Because the sweep forms broad task-dependent plateaus under finite evaluation variance, we do not treat a single point-best $\sigma_{\max}$ or small within-plateau score differences as meaningful checkpoint ordering. Appendix N records a negative target-view ablation: perturbed-history $\to$ original-future denoising is not a sufficient closed-loop fix, so the retained empirical branch is full-sequence perturbed-target training.

1.4 Contributions

C1 — Multi-stage diagnostic principle. We operationalize visual robustness for JEPA world-model control as a multi-stage diagnostic rather than an encoder-invariance score alone. Encoder perturbation geometry measures whether visual noise enters and reshapes the latent state; action-conditioned predictive consistency measures whether that shift changes predicted futures and candidate costs; the discriminability countercondition prevents contraction from erasing action-relevant distinctions.

C2 — Planner link and controlled evidence. Under a Lipschitz cost readout, bounded rollout disagreement bounds candidate-cost drift on a fixed candidate set. For a once-sampled candidate pool, the top-1 flip probability is controlled by two explicit terms: ACPC tails and clean candidate-margin tails. A local Gaussian linearization shows that the measured ACPC basin radius estimates action-conditioned encoder–predictor sensitivity $\|J_G J_E\|_F$, not encoder invariance alone. These results support the diagnostics while leaving adaptive CEM resampling, repeated replanning, and closed-loop trajectories to empirical evaluation. We use a fixed Gaussian-noise protocol to probe the encoder–predictor–planner chain on LeWM, with PLDM as a bounded second-family replication check.

C3 — Three-seed diagnostic evidence chain. On fixed checkpoints, training-seed robustness establishes the matched-Gaussian behavior; selector-baseline-qualified plateau localization tests whether fixed readouts stay near held-out seed plateaus; margin-conditioned fixed-pool action flips check the clean-margin side of the sampled-pool link; and local task-feature proxy margins test selective contraction. Larger LeWM/PLDM grids, blur/resize scope checks, ablations, and implementation key maps are kept in the appendix so the main text follows this evidence chain.

2 Related Work

2.1 Latent prediction and JEPA world models

JEPA [23] predicts in latent space rather than reconstructing pixels; I-JEPA [2] and the V-JEPA family [4, 3, 27] extend this idea to images, video, and dense physical-world representations. LeWM [26], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [8]. PLDM [33, 34], via `stable-worldmodel` [25], supplies the second method family in Appendix I. LeJEPA theory [19] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [41, 36, 45]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [39] derive closed-form SSL solutions showing that joint embedding imposes weaker alignment requirements than reconstruction when irrelevant features have large magnitude, while Littwin et al. [24] analyze deep-linear self-distillation dynamics and a JEPA bias toward high-influence predictive features rather than merely high-variance features. These analyses are representation-level and do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. ACPC is complementary: it asks whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable. Seq-JEPA [12] addresses a related invariance–equivariance tension architecturally. DrQ/DrQ-v2 [21, 43], SODA/DMC-GB [16], DreamerV3 [14], TD-MPC2 [15], ViGMO [28], N-JEPA [29], variational VJEPA [17], and US-JEPA [30] provide visual-robustness and noisy-environment context. They are not trained as competing baselines in this study; they define important future comparisons for method or benchmark claims.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [11], DBC [44], bisimulation-regularized MPC [32], and Bisim-JEPA [38] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [13, 40] and VIBR [7] likewise emphasize downstream control consequences. Wang et al. [42] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [6] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

2.4 Representation diagnostics and collapse analysis

We use standard representation diagnostics: effective rank [31, 18, 37], CKA [20], linear probes [1], and anti-collapse context from VICReg [5]. RankMe [10] motivates treating representation geometry as a diagnostic rather than as a release criterion. The individual metrics are standard; our contribution is the protocol that ties them to closed-loop Gaussian-noise evaluation and ACPC-style paired checks. A compact boundary table is provided in Appendix B.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while action-, transition-, or cost-distinct states should remain separable. Section 3.2 then separates the main descriptive diagnostics from paired ACPC probes.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation–action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length-k prefix of this sequence after re-indexing time at t, and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \qquad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \tag{2}$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\mathrm{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^{H} \alpha_k\, d\big(\Pi(\hat{z}_{t+k}),\, \Pi(\hat{\tilde{z}}_{t+k})\big), \qquad \alpha_k \geq 0. \tag{3}$$

For the theoretical statements below, we use the induced horizon metric

$$d_H\big((u_1, \ldots, u_H), (v_1, \ldots, v_H)\big) = \sum_{k=1}^{H} \alpha_k d(u_k, v_k), \tag{4}$$

so that Equation (3) is exactly the clean/perturbed rollout-readout distance under the shared action sequence. In experiments we instantiate this with the L2 distance over rollout tokens used by R_F. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta_{\mathrm{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y\big(Y^H(s_i, \mathbf{a}),\, Y^H(s_j, \mathbf{a})\big) > m, \tag{5}$$

the representation and predictions should keep them separable,

$$d\big(\Psi(z_i),\, \Psi(z_j)\big) > m', \tag{6}$$

where Ψ is a discriminability readout, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (3)–(6) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\mathrm{ACPC}_H \to 0$ by collapsing the latent would violate (6), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The main paper uses one fixed ACPC instance. For the Gaussian-noise basin diagnostic, Π is the identity readout on the model inference/cost latent rollout, d is Euclidean L2 over rollout tokens, and $H = 8$. The reported quantities are R_E, the median same-state encoder-history spread normalized by the original-view nearest-neighbor L2 scale, and R_F, the median same-state rollout-latent spread normalized by the unperturbed transition L2 reference over the same horizon. R_F/R_E is a descriptive contraction ratio. This empirical R_F is a distributional, recorded-action localization proxy. It should not be read as an estimate of the uniform ϵ in the fixed-candidate stability corollary, which requires a bound over every candidate in a shared set $\mathcal{A}$. A compact shared-candidate PCC/CRA/MAF table in Section 5.5 checks the downstream direction of this link; the full LeWM/PLDM readout table remains in Appendix L.

Thus R_E and R_F answer different questions. R_E measures whether the visual perturbation changes the encoded neighborhood; R_F measures whether that change survives action-conditioned rollout in coordinates used for planning. Robust checkpoints in this Gaussian-noise study often reduce both; the diagnostic point is to interpret encoder contraction through its downstream predictive effect and the accompanying discriminability guard.

The main text is organized around three necessary questions rather than a long metric list. Table 1 names the readout used for each question; auxiliary ADM/SPRR, latent-noise, and full representation diagnostics are kept in the appendix.

Table 1: Core readout map used in the main text. Each row answers one necessary question in the ACPC story; implementation keys and auxiliary metrics are deferred to Appendix C.10.

Question	Main readout	Role
Does the behavioral effect survive seed variance?	closed-loop success/drop, training-seed mean $\pm$ std	establishes the Gaussian robustness phenomenon before interpreting internals
Does same-state noise change planning-relevant prediction?	R_F plus PCC/CRA/MAF and margin-conditioned flips as one rollout/cost family	localizes whether clean/noisy views induce consistent action-conditioned rollouts, candidate costs, and fixed-pool action choices
Was contraction selective?	local task-feature proxy margin plus transition/ID proxy checks	checks that action-relevant distinctions are not erased while nuisance views contract

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g, define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \ldots, \Pi(\hat{z}_{t+H}), g)\,, \tag{7}$$

where J is the planner's cost readout on the rollout-readout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (Encoder shift as one route to rollout disagreement)**.** *Fix an action sequence* $\mathbf{a}$ *and write*

$$G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$$

for the rollout-readout map. If $G_{\mathbf{a}}$ is locally L_G-Lipschitz between $E_\theta(h)$ and $E_\theta(\tilde{h})$ under encoder metric d_Z and rollout-readout metric d_H, then

$$d_H\big(G_{\mathbf{a}}(E_\theta(h)), G_{\mathbf{a}}(E_\theta(\tilde{h}))\big) \le L_G\, d_Z\big(E_\theta(h), E_\theta(\tilde{h})\big)\,.$$

Proof. This is the local Lipschitz condition applied to the two encoded histories. □

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by $G_{\mathbf{a}}$, and small shifts can be amplified by the predictor or cost readout. This is why the empirical analysis reports both encoder radii R_E and rollout radii R_F.

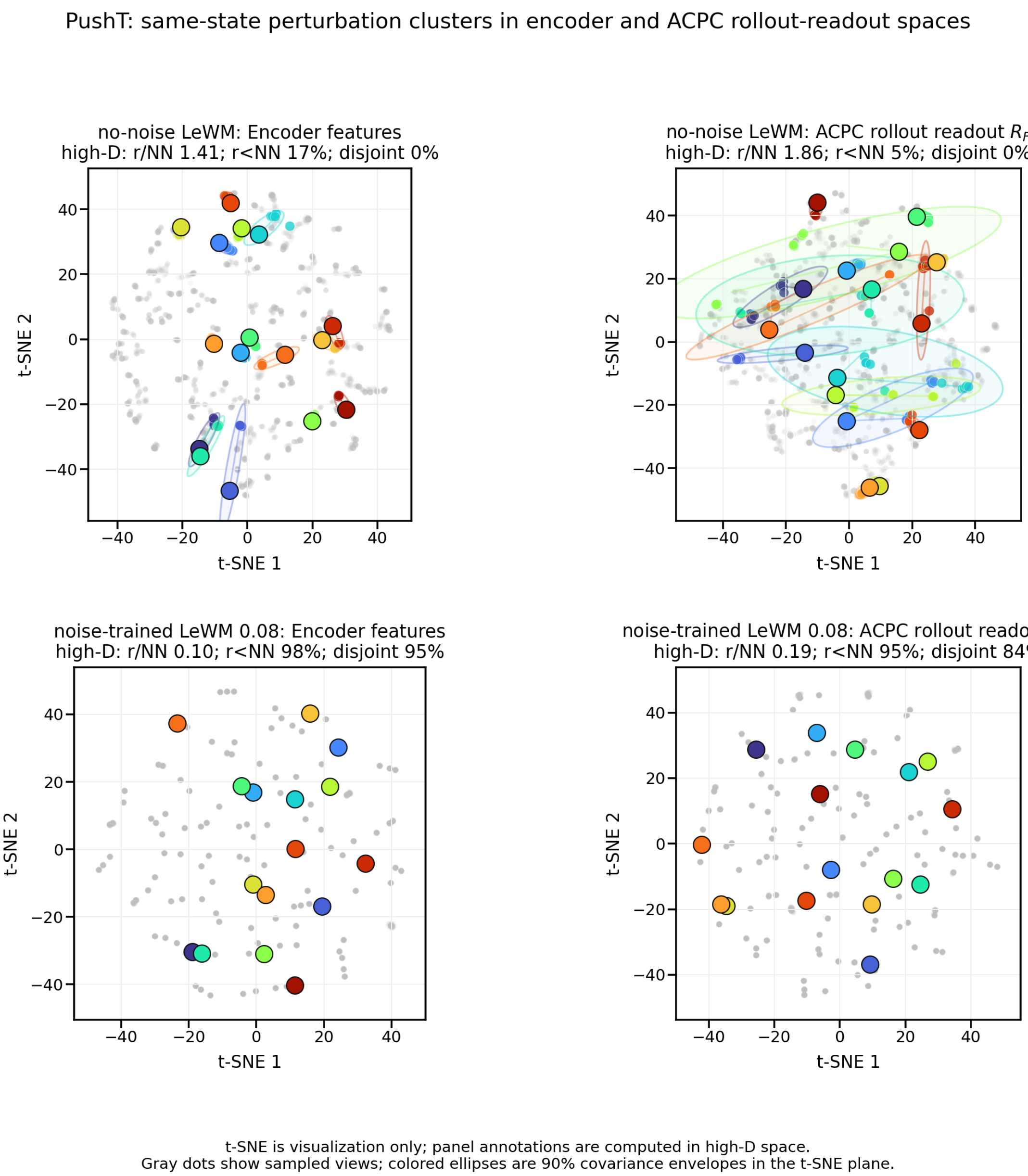

Figure 1: Empirical ACPC-basin visualization used as the paper's main visual reference. Top: no-noise-trained LeWM on PushT; bottom: noise-trained LeWM at $\sigma_{\max} = 0.08$. Left: encoder-history features; right: ACPC rollout readout R_F under the same recorded action sequence. t-SNE coordinates and envelopes are visualization only; small panel summaries report original-space local statistics. The quantitative multi-task basin summary is reported in Table 7.

Proposition 2 (ACPC controls candidate-cost drift). *Assume that J is locally L_J-Lipschitz on the clean/perturbed rollout-readout neighborhood evaluated by the fixed candidate set under metric d_H. If*

$$d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))\big) \le \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \le L_J \epsilon.$$

Proof. The inequality is the local Lipschitz condition applied directly to the clean and perturbed rollout-readout sequences. □

In the reported basin diagnostic, this condition is instantiated with the L2 rollout-token distance underlying R_F; other downstream readouts are treated only as exploratory diagnostics.

Proposition 3 (Candidate top-1 stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \ge C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \ge \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. □

Corollary 1 (ACPC-margin condition for candidate stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be a shared candidate set and assume the cost readout J is locally L_J-Lipschitz as in Proposition 2. If every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ between the clean and perturbed branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le L_J\epsilon$ for every candidate. Applying Proposition 3 with $\eta = L_J\epsilon$ gives the result. □

The corollary gives a formal link between ACPC and planning without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. Candidate-ranking agreement and margin-conditioned action flips are downstream readouts of this chain. The result is limited to a fixed candidate set; CEM resampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

3.4 Sampled-pool stability and Gaussian sensitivity

The fixed-candidate statement is the deterministic core. A single MPC or CEM iteration, however, evaluates a sampled candidate pool. The next bound keeps the same diagnostic scope while exposing the two terms the empirical readouts estimate: ACPC tails and clean-margin tails.

Theorem 1 (Sampled-pool ACPC stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool, and use deterministic tie-breaking for* $\arg\min$. *For each sampled candidate define*

$$D_j = d_H\big(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}^j)), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}^j))\big).$$

Assume J is locally L_J-Lipschitz on the evaluated rollout-readout neighborhood and that a single draw from q satisfies

$$\Pr_{\mathbf{a}\sim q}[D(\mathbf{a}) > \epsilon] \le \delta.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean branch top-1/top-2 margin inside the sampled pool. Then

$$\Pr_{\mathcal{A}}\left[\arg\min_j C_h(\mathbf{a}^j, g) \neq \arg\min_j C_{\tilde{h}}(\mathbf{a}^j, g)\right] \le K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \le 2L_J\epsilon].$$

Theorem 1 is not a convergence theorem for adaptive multi-round CEM. It controls the instability of one sampled candidate pool. The first term is the probability that at least one candidate has a large clean/noisy rollout discrepancy; the second term is the probability that the clean candidate pool has too small a top-1 margin. This is why the shared-candidate diagnostics are reported as a family rather than a single scalar: ACPC-H and PCC estimate rollout/cost drift, CRA estimates ranking stability, and MAF estimates the high-margin flip failure term. A repeated replanning episode simply accumulates such risks unless additional assumptions are made.

Corollary 2 (Replanning union bound). *If replanning step t has sampled-pool flip-probability bound*

$$r_t = K_t\delta_t + \Pr[\Delta_{\mathcal{A}_t} \leq 2L_J\epsilon_t],$$

then over T replanning steps,

$$\Pr[\exists t \leq T: \text{ the sampled-pool top-1 flips at step } t] \leq \sum_{t=1}^{T} r_t.$$

Proposition 4 (Finite-sample tail calibration for paired diagnostics). *Fix the diagnostic sampling distribution used to draw independent paired clean/noisy histories and candidate action sequences. Let*

$$X_i = \mathbf{1}[D_i > \epsilon], \qquad i = 1, \dots, n,$$

where D_i is the paired ACPC rollout-readout discrepancy for diagnostic item i. Let $p_\epsilon = \Pr[D > \epsilon]$ and $\hat{p}_\epsilon = n^{-1}\sum_i X_i$. Then, with probability at least $1-\delta$ over the diagnostic sample,

$$p_\epsilon \leq \hat{p}_\epsilon + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

The same one-sided calibration applies to any bounded Bernoulli readout computed from paired candidate costs, such as a margin-conditioned action-flip indicator.

This calibration is with respect to the released diagnostic sampling distribution, its independent-sample idealization, and the finite candidate construction. It does not convert an empirical median, quantile, or flip-rate table into a uniform bound over all CEM samples or all closed-loop states; when nearby dataset windows are dependent, we use it only as calibration.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 5 (Local Gaussian ACPC sensitivity). *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ,*

$$G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi\left[\|G_{\mathbf{a}}(E_\theta(o+\xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2\right] = \sigma^2 \|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}}J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while rollout readouts for action-relevant state differences remain separated. The local task-feature proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m],$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic calibration rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure—rollout-disagreement tails and clean candidate-margin tails—and explain why the Gaussian basin radius is an action-conditioned encoder–predictor sensitivity measurement. They do not assert that a low empirical median radius is a uniform bound for adaptive CEM, repeated replanning, or environment-feedback stability.

Selective ACPC pseudo-metric. For a nonempty fixed candidate family $\mathcal{A}$, the stability condition can be summarized by

$$D^{\mathcal{A}}_{\mathrm{ACPC}}(h,\tilde{h}) = \max_{\mathbf{a}\in\mathcal{A}} d_H\left(\Pi(F^{1:H}_\theta(E_\theta(h),\mathbf{a})), \Pi(F^{1:H}_\theta(E_\theta(\tilde{h}),\mathbf{a}))\right). \tag{8}$$

When d_H is a metric (or a pseudo-metric, if some horizon weights vanish), $D^{\mathcal{A}}_{\mathrm{ACPC}}$ is a pseudo-metric on histories after passing through the encoder, predictor, and readout: distinct histories can have zero distance if all candidate rollouts are readout-equivalent. This follows because each map $h \mapsto \Pi(F^{1:H}_\theta(E_\theta(h),\mathbf{a}))$ pulls back d_H to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small $D^{\mathcal{A}}_{\mathrm{ACPC}}$ for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task readouts differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

Corollary 3 (Fixed-candidate predictive stability)**.** *Assume $D^{\mathcal{A}}_{\mathrm{ACPC}}(h,\tilde{h}) \le \epsilon$ for a same-state visual-perturbation pair and that J is locally L_J-Lipschitz on the rollout readout. If the clean candidate margin satisfies $\Delta > 2L_J\epsilon$, then the clean and perturbed branches select the same top-1 candidate from the fixed set $\mathcal{A}$.*

Proof. By definition of $D^{\mathcal{A}}_{\mathrm{ACPC}}$, every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ. Proposition 2 bounds the cost drift of every candidate by $L_J\epsilon$, and Proposition 3 gives top-1 stability when $\Delta > 2L_J\epsilon$. □

The statement becomes selective only when it is paired with a discriminability condition such as Equation (6). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_\theta(h)$ and $E_\theta(\tilde{h})$ far apart while leaving $\Pi(F^{1:H}_\theta(\cdot,\mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

Proposition 6 (ACPC alone permits collapse)**.** *There exist encoder–predictor pairs for which* $\mathrm{ACPC}_H = 0$ *for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

Proof. Let $E_\theta(h) = z_0$ for every history and let $F^k_\theta(z_0,\mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and perturbed branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (6). □

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of the transition-resolution and inverse-dynamics probes in the main diagnostics, and it is the failure exposed by the heteroscedastic-loss ablation in Appendix G. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned rollout readouts and candidate costs while preserving action-relevant separations.

4 Study protocol

4.1 Models and noise intervention

LeWM [26] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization; inference uses CEM for latent-space MPC. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\mathrm{Uniform}(0, \sigma_{\max})$. Because the target future frame is also encoded from the noised sequence, this intervention is not an original-target denoising objective. We sweep $\sigma_{\max} \in \{0.01, \dots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal Gaussian noise is retained as an auxiliary stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean ± population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Table 4. The runs use a single NVIDIA H800 (80 GB).

The main text separates four must-read results from reproducibility material. First, Table 4 reports training-seed mean/std for closed-loop Gaussian robustness. Second, Tables 5 and 6 evaluate one ACPC/PCC/CRA/MAF rule fixed on seed 3072 and then applied to held-out training seeds 3073/3074 over the full 3-seed × 4-task × 9-checkpoint LeWM grid. Third, Table 9 audits the sampled-pool clean-margin term through margin-conditioned top-1 flips. Fourth, Table 11 reports local task-feature proxy margin pass-rates for the no-noise and fixed high-noise endpoints. Unseen-stressor scores, PLDM replication, target-view and heteroscedastic-loss ablations, latent-noise probes, exact key names, and full diagnostic tables are kept as scope checks or reproducibility appendices.

PLDM replication is reported in Appendix I.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; point-best noise-level ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 2 reports LeWM-base success rates under unperturbed and noised evaluation (mean ± population std across three evaluation seeds × 100 evaluations).

Table 2: LeWM-base under test-time Gaussian-noise settings (mean ± population std; 3 evaluation seeds × 100 evaluations). Drop is clean success minus observation-noise 0.08 success.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	94.00 ± 3.56	72.33 ± 5.79	65.67 ± 1.25	50.00 ± 1.41	28.33
PushT	86.33 ± 2.36	17.00 ± 1.41	4.33 ± 1.25	4.67 ± 2.05	82.00
Reacher	58.67 ± 1.25	33.67 ± 2.87	18.33 ± 2.05	15.00 ± 2.16	40.33
Cube	66.67 ± 2.62	48.67 ± 6.85	47.00 ± 6.38	46.33 ± 3.68	19.67

LeWM-base is strong on unperturbed images (especially TwoRoom and PushT), but observation noise alone already produces large closed-loop drops: PushT loses 82.00 pts at observation-noise 0.08, Reacher 40.33 pts, TwoRoom 28.33 pts, and Cube 19.67 pts. The observation+goal column shows the stronger full-input-stream Gaussian-noise stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: Cube degrades least, while PushT degrades most sharply, consistent with contact-heavy continuous control being most sensitive to visual precision.

PLDM provides a second-family boundary on this failure mode. Without noise augmentation, PLDM drops sharply on PushT ($75.33\% \pm 3.68$ unperturbed to $18.33\% \pm 2.05$ at observation-noise 0.08, 57.00 pt) and TwoRoom ($96.67\% \pm 1.70$ to $67.00\% \pm 2.16$, 29.67 pt), while Reacher and Cube have much smaller no-noise-training gaps (2.33 and 4.33 pt; Table 22). PLDM Reacher is therefore a useful boundary case: its no-noise baseline is already high, so the LeWM Reacher clean lift below is read as a LeWM fixed-epoch stabilization phenomenon rather than a universal property of noise training. On the PLDM tasks with substantial cliffs, noise-trained rows recover observation-noise performance; Cube remains weakly responsive (Appendix I). These rows provide replication and boundary evidence for the same diagnostic story.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Table 3 summarizes the sweep points needed for the main argument; the complete 8-level tables are in Appendix C.4. All entries use the unified $n = 3$ seeds (42/43/44) $\times$ 100 trajectories protocol.

Table 3: LeWM noise-sweep summary. Values are success rate mean $\pm$ population std across evaluation seeds. The two point-best columns may refer to different training-noise levels; they summarize the sweep envelope, while the full sweep in Figure 2 and Appendix C.4 provides the grid context.

Task	unpert. base	obs 0.08 base	best clean row	best obs0.08 row	Main reading
TwoRoom	94.00 ± 3.56	65.67 ± 1.25	98.33 ± 0.47 (0.08)	97.67 ± 0.94 (0.08)	large recovery; visually redundant regime
PushT	86.33 ± 2.36	4.33 ± 1.25	89.67 ± 1.70 (0.03)	89.00 ± 4.32 (0.08)	large recovery; contact-sensitive state
Reacher	58.67 ± 1.25	18.33 ± 2.05	86.00 ± 2.94 (0.06)	84.67 ± 5.19 (0.07)	recovery plus fixed-epoch clean lift
Cube	66.67 ± 2.62	47.00 ± 6.38	69.33 ± 0.47 (0.01)	68.33 ± 2.49 (0.03)	weaker recovery signal

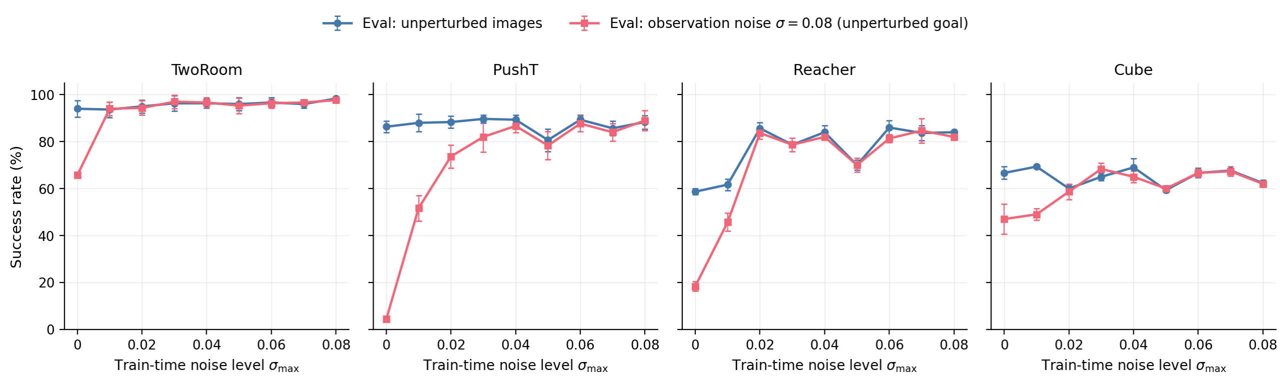

Figure 2: Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document the intervention grid and the representative recovered checkpoints used for downstream diagnostics.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The diagnostic table supports this reading: Reacher does not show a resolution collapse under the fixed $\sigma_{\max} = 0.08$ diagnostic checkpoint, with effective rank and transition resolution flat or slightly improved, while multi-step rollout drift drops sharply ($15.17 \rightarrow 0.36$). Thus the same intervention that contracts same-state noisy rollouts coincides with more stable clean action-conditioned prediction at epoch 10. This is a diagnostic observation, not a claim about asymptotic training behavior. The three-training-seed replication below makes the endpoint less dependent on seed 3072, but longer learning curves would still be needed to separate regularization from optimization timing.

The completed Gaussian training-seed replication strengthens the closed-loop part of the claim. It repeats the full Gaussian sweep for LeWM seeds 3073 and 3074 and combines them with the canonical seed-3072 sweep. Table 4 summarizes the observation-only Gaussian endpoint across all three training seeds.

Table 4: Three-training-seed Gaussian replication for LeWM. Values are mean ± population std across training seeds 3072/3073/3074; each seed point is itself the mean over evaluation seeds 42/43/44 with 100 trajectories per eval seed. The endpoint is observation-only Gaussian evaluation at 0.08 with the goal image unperturbed. Regret is best obs 0.08 minus the fixed $\sigma_{\max} = 0.08$ endpoint; the best-$\sigma_{\max}$ range shows the task/seed-dependent plateau.

Task	base obs 0.08	std0.08 obs 0.08	gain	best obs 0.08	regret	best $\sigma_{\max}$ range
TwoRoom	68.78 ± 2.46	97.11 ± 0.57	28.33 ± 2.68	97.67 ± 0.27	0.56 ± 0.79	0.05–0.08
PushT	7.22 ± 5.81	85.78 ± 3.45	78.56 ± 5.18	87.67 ± 0.98	1.89 ± 2.67	0.04–0.08
Reacher	18.22 ± 0.42	81.55 ± 0.88	63.34 ± 1.25	83.78 ± 0.63	2.22 ± 0.87	0.05–0.07
Cube	43.11 ± 3.27	62.56 ± 1.55	19.45 ± 4.53	66.44 ± 1.34	3.89 ± 2.20	0.03–0.05

Across three training seeds, the no-noise checkpoints retain large observation-noise cliffs, while the $\sigma_{\max} = 0.08$ endpoint recovers most of the best observed sweep performance. PushT remains the sharpest stress case, improving by 78.56 ± 5.18 points at the fixed high-noise endpoint; Reacher improves by 63.34 ± 1.25 points. The nonzero regret and task-dependent best-$\sigma_{\max}$ ranges support plateau wording rather than a universal optimum at exactly $\sigma_{\max} = 0.08$.

A three-seed blur/resize score check is reported in Appendix K. It is a deliberately narrow scope check on severe non-Gaussian stressors, not part of the three main results. The diagnostic validation below instead uses the complete Gaussian full-grid run over all three LeWM training seeds.

5.3 Three-seed fixed-rule diagnostic validation

We fix one diagnostic rule on the seed-3072 development grid before evaluating independent training seeds: among nonzero training-noise checkpoints, rank ACPC-H/transition lower, PCC lower, CRA higher, and MAF lower, then select the lowest aggregate rank. The selector uses paired rollout/candidate-cost readouts only; no closed-loop success enters the score. It is then applied unchanged to held-out training seeds 3073/3074, giving 8 held-out task-seed blocks and 64 candidate checkpoints; the full three-seed grid gives 12 blocks and 96 candidates.

Table 5: Development/held-out split for the fixed diagnostic rule. Seed 3072 freezes metric computation and the aggregate-rank selector; seeds 3073/3074 are held-out training seeds. Bootstrap CI is over task-seed blocks.

Split	blocks	candidates	within 5pp	regret to best	95% CI
development 3072	4	32	3/4	2.33 ± 3.49	$[0.00, 6.25]$
held-out 3073/3074	8	64	7/8	2.21 ± 1.83	$[1.04, 3.54]$
all seeds	12	96	10/12	2.25 ± 2.51	$[1.00, 3.75]$

Table 6: Three-seed fixed-rule diagnostic validation on the full LeWM Gaussian grid. Within-5pp counts how often the selected checkpoint is within five percentage points of the closed-loop best observation-noise 0.08 checkpoint in the same task/seed block. Regret is best minus selected observation-noise 0.08 success.

Task	exact best	within 5pp	regret to best	top-2 overlap
TwoRoom	1/3	3/3	0.56 ± 0.79	0.67/2
PushT	1/3	2/3	2.22 ± 2.47	0.67/2
Reacher	0/3	3/3	1.67 ± 0.94	1.67/2
Cube	0/3	2/3	4.56 ± 3.00	0.00/2
All	2/12	10/12	2.25 ± 2.51	0.75/2

The fixed rule is not a replacement for closed-loop evaluation, and its purpose is narrower than checkpoint selection: it tests whether ACPC-family readouts localize broad plateaus on held-out seeds. Across the 96 nonzero checkpoint rows, signed ACPC-H/transition has Spearman $\rho = 0.41$ with observation-noise 0.08 success and $\rho = 0.62$ with reduced drop; PCC gives $\rho = 0.38$ and 0.60 on the same two targets. The held-out audit is strongest on TwoRoom and Reacher, useful but imperfect on PushT, and weakest on Cube. The clearest miss is Cube seed 3072: the fixed rule chooses $\sigma_{\max} = 0.05$ with 60.00% observation-noise 0.08 success, while the closed-loop best is $\sigma_{\max} = 0.03$ at 68.33% (regret 8.33 pp). PushT seed 3073 is the other beyond-5pp miss (81.00% selected at $\sigma_{\max} = 0.08$ versus 86.67% best at $\sigma_{\max} = 0.04$). A selector-baseline audit in Appendix M shows the same boundary: the aggregate rule is comparable to fixed $\sigma_{\max} = 0.08$ and single-metric selectors on broad Gaussian plateaus, but better than exact random nonzero-std selection. We therefore use the fixed rule to explain plateau structure and failure cases, while closed-loop scores remain the authority for final claims.

5.4 Paired Gaussian-noise ACPC basin diagnostic

We compute an eval-only ACPC basin diagnostic on the same 36 LeWM epoch-10 checkpoints used in the sweep. For each checkpoint we sample 100 fixed dataset windows and create eight paired same-state views using Gaussian pixel noise with std $\in \{0.01, \ldots, 0.08\}$, matching the training sweep's noise family. Each original/noised branch is encoded and then rolled out under the same recorded action sequence for horizon $H = 8$. The reported prediction readout is the identity on the resulting inference/cost latent rollout sequence, with L2 distance over rollout tokens. The released source and rendering script are listed in Appendix C.10. The diagnostic intentionally excludes blur/resize so that the ACPC evidence is not confounded by train/eval perturbation-family mismatch.

For each checkpoint, let R_E be the median pairwise spread among the original and noised encoder-history views, normalized by the original-view local NN L2 scale, and let R_F be the median pairwise spread among their 8-step rollout latents, normalized by the unperturbed transition L2 reference over the same horizon. Lower R_F means the same-state perturbation views induce a tighter action-conditioned predictive basin; R_F/R_E is a descriptive contraction ratio rather than a success predictor.

This comparison localizes where the intervention changed same-state perturbation behavior. The compact rows summarize the high-noise regime visible in closed-loop evaluation; the fixed-rule validation in Table 6 audits plateau localization across the full three-seed grid. Appendix C.5 reports the full basin grid, and Section 5.5 reports a compact shared-candidate downstream check.

Table 7: LeWM Gaussian-noise ACPC-basin proxy: no-noise baseline vs. representative high-noise grid rows. R_E is normalized encoder-view spread and R_F is normalized action-conditioned rollout-view spread under the same action sequence. 8-step drift is the standard 8-step rollout-drift diagnostic at the same checkpoint; it is a rollout-space companion to R_F, not a discriminability guard. Drop is clean success minus observation-noise 0.08 success.

Task	checkpoint	obs 0.08	drop	R_E	R_F	8-step drift
TwoRoom	base	65.67	28.33	3.538	0.785	18.62
TwoRoom	noise 0.08	97.67	0.67	0.235	0.028	0.66
PushT	base	4.33	82.00	1.727	1.543	18.65
PushT	noise 0.08	89.00	−0.67	0.137	0.088	1.06
Reacher	base	18.33	40.33	4.238	1.012	15.17
Reacher	noise 0.07	84.67	−1.00	0.135	0.027	0.40
Cube	base	47.00	19.67	1.343	0.957	20.20
Cube	noise 0.03	68.33	−3.33	0.238	0.115	7.53

The compact rows quantify the characteristic basin movement illustrated in Figure 1: R_F is much smaller than at the no-noise baseline on the tasks with large recovery, with PushT decreasing from 1.543 to 0.088 and TwoRoom from 0.785 to 0.028. The companion 8-step rollout-drift column gives the same rollout-space localization in the standard diagnostic suite: it drops sharply on TwoRoom, PushT, and Reacher, while Cube's weaker recovery leaves a larger residual drift. Appendix C.8 adds a projection-free local-atlas check. The comparison localizes a joint contraction pattern after the closed-loop sweep: recovered checkpoints usually shrink both R_E and R_F, with 8-step drift as a rollout-stability companion; the anti-collapse evidence comes from the downstream discriminability checks.

The basin contraction is not treated as sufficient evidence of robustness. It is paired with the downstream readouts in Table 8 and the discriminability checks in Table 10: the former checks that shared-candidate costs and rankings move in the expected direction, while the latter rules out the trivial explanation that low R_F was obtained by erasing action-relevant state distinctions.

5.5 Downstream shared-candidate readouts move consistently

The fixed-candidate stability argument predicts that smaller same-action rollout disagreement should reduce downstream candidate-cost drift and ranking instability, subject to the stated margin assumptions. Table 8 reports a compact LeWM view from the released paired-diagnostic run using the primary observation-noise endpoint, horizon $H = 8$, and shared candidate-action sets. These representative rows connect the basin diagnostic to planner-side quantities; Table 6 gives the separate three-seed selection readout.

Table 10: Compact representation discriminability checks: LeWM-base → the fixed $\sigma_{\max} = 0.08$ noise-trained checkpoint. These checks are paired with the local task-feature proxy margin pass-rate in Table 11.

Task	$\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2
TwoRoom	0.08	47.60 → 37.69	0.7216 → 0.6621	0.2889 → 0.1419
PushT	0.08	76.42 → 78.08	0.3015 → 0.2789	0.7739 → 0.7647
Reacher	0.08	61.04 → 66.20	0.3704 → 0.3831	0.1621 → 0.1767
Cube	0.08	73.25 → 74.97	0.4847 → 0.5085	0.6657 → 0.6342

Table 8: Compact LeWM shared-candidate readouts at the primary observation-noise endpoint. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA. The full LeWM/PLDM table is in Appendix L.

Task	obs-noise success	ACPC-H/trans.	PCC	CRA	MAF
TwoRoom	65.67 → 97.67	1.178 → 0.041	42.0 → 1.0	0.364 → 0.999	0.81 → 0.04
PushT	4.33 → 87.67	2.625 → 0.146	56.6 → 2.6	0.412 → 0.997	0.98 → 0.08
Reacher	18.33 → 83.67	2.092 → 0.029	72.2 → 0.5	0.278 → 1.000	0.91 → 0.01
Cube	47.00 → 67.33	1.979 → 0.058	63.7 → 1.3	0.274 → 0.999	0.89 → 0.00

The table strengthens the interpretation of R_F as a control-facing localization signal: on all four LeWM tasks, smaller paired rollout disagreement is accompanied by lower candidate-cost drift, higher candidate-ranking agreement, and fewer high-margin action flips. This is the empirical counterpart of Theorem 1: ACPC-H and PCC estimate the drift/tail side of the bound, CRA checks whether sampled candidate rankings remain aligned, and MAF estimates the high-margin flip failure term after margin filtering. The fixed-rule validation in Tables 5 and 6 uses the same theory-matched readout family for no-retraining plateau localization, while closed-loop evaluation remains the final authority.

Because the sampled-pool link depends on the clean top-1/top-2 candidate margin, we also run a sample-level margin-conditioned action-flip audit on the same fixed checkpoints. For each sampled state we record the clean candidate margin and whether the clean and noisy branches choose different top-1 candidates from the same 65-action candidate set. Table 9 reports flip rates after filtering to the top clean-margin quartile within each checkpoint; the all-sample column is shown as context. The recovered endpoint removes high-margin flips in all 12 task–training-seed blocks at this threshold. This is a finite fixed-pool calibration of the clean-margin term, not a closed-loop CEM guarantee.

Table 9: Sample-level margin-conditioned action-flip audit over training seeds 3072/3073/3074. The q75 columns filter to the top clean-margin quartile within each checkpoint (25 samples per task–seed–endpoint); all-sample columns use all 100 samples. Lower flip rate is better.

Task	q75 flip base	q75 flip std0.08	all flip base	all flip std0.08
TwoRoom	0.67 ± 0.07	0.00 ± 0.00	0.80 ± 0.03	0.03 ± 0.00
PushT	0.92 ± 0.06	0.00 ± 0.00	0.92 ± 0.03	0.03 ± 0.02
Reacher	0.79 ± 0.02	0.00 ± 0.00	0.88 ± 0.03	0.01 ± 0.00
Cube	0.84 ± 0.09	0.00 ± 0.00	0.92 ± 0.03	0.00 ± 0.01

5.6 Selective consistency and failure checks

Low ACPC would be uninformative if it came from collapsed representations. Because 8-step drift already appears in the ACPC-basin section as rollout-space localization, Table 10 keeps only the compact representation checks: effective rank, transition-resolution ratio, and inverse-dynamics probe R^2. Table 11 then adds the local task-feature proxy margin readout.

The representation checks are paired with a local task-feature proxy margin pass-rate. For each task, we use a state-proxy factor available in the logged stream: TwoRoom agent position/doorway-side coordinate, PushT T-block pose, Reacher target/end-effector relation proxy, and Cube cube-pose/goal-relation proxy. Instead of using globally far state pairs, each anchor searches within the closest 35% state-distance neighborhood and selects the neighbor with the largest task-feature contrast. A sample passes if this local proxy-different clean rollout distance is larger than its same-state clean/noisy rollout radius. This is still a programmatic state-proxy audit, not an oracle hand-labeled semantic proof, but it is a stricter near-boundary check than the earlier median-distance sanity pass. Table 11 reports the three-training-seed mean/std for the no-noise baseline and the fixed $\sigma_{\max} = 0.08$ endpoint.

Table 11: Local task-feature proxy selective ACPC study over training seeds 3072/3073/3074. Pass-rate is the fraction whose local proxy-different clean rollout distance exceeds its same-state noisy radius under the same-action rollout readout. Pairs are selected inside the closest 35% state-distance neighborhood by maximum task-feature contrast.

Task	std	pass-rate	same radius	proxy diff	margin	local state
TwoRoom	0.0	0.59 ± 0.05	17.45 ± 0.65	18.51 ± 0.25	0.98 ± 0.49	0.93
TwoRoom	0.08	1.00 ± 0.00	0.56 ± 0.02	17.98 ± 0.11	17.40 ± 0.19	0.93
PushT	0.0	0.54 ± 0.17	18.39 ± 1.24	18.96 ± 0.16	0.56 ± 1.51	2.45
PushT	0.08	1.00 ± 0.00	0.70 ± 0.06	19.10 ± 0.07	18.18 ± 0.07	2.45
Reacher	0.0	0.81 ± 0.02	15.44 ± 0.51	19.51 ± 0.02	4.08 ± 0.44	2.84
Reacher	0.08	1.00 ± 0.00	0.27 ± 0.02	19.41 ± 0.15	19.10 ± 0.12	2.84
Cube	0.0	0.53 ± 0.05	19.33 ± 0.15	19.47 ± 0.13	0.29 ± 0.37	5.90
Cube	0.08	1.00 ± 0.00	0.49 ± 0.05	19.55 ± 0.06	19.03 ± 0.14	5.90

The local task-feature proxy margin supplies the missing selective half of ACPC more directly than rank or ID probes alone, while staying below an oracle semantic-label claim. The high-noise endpoint keeps local proxy-different rollouts separated while shrinking same-state noisy rollouts on all four tasks; the no-noise baselines are weaker under this stricter local-neighbor audit, especially PushT and Cube. Each task/endpoint aggregates 100 sampled anchors per training seed and three training seeds. Treating these local pairs as independent only for calibration, Hoeffding gives $p \geq \hat{p} - \sqrt{\log(1/\alpha)/(2n)}$; for the rounded high-noise pass-rates $\hat{p} \approx 1.00$, $n \approx 300$, and $\alpha = 0.05$, the lower confidence bound is about 0.929. Nearby dataset windows may be dependent, so we use this as a finite-sample calibration rather than a dataset-wide formal guarantee. The heteroscedastic-loss ablation in Appendix G remains the failure case that motivates the guard: PushT transition L2 falls from 0.3015 to 0.1023, the ID probe falls from 0.7739 to 0.2678, and unperturbed success collapses to 13.33%. The longer representation, rollout-drift, latent-noise, and task-by-task mechanism diagnostics remain in Appendix D–H.

5.7 Bounded unseen-stressor scope check

The main claim remains matched Gaussian-noise robustness. After freezing the Gaussian grid, we also ran a strongest-severity unseen-stressor check to test whether the diagnostic direction is specific outside the training perturbation family. Table 12 shows a split pattern. TwoRoom and Reacher exhibit clear positive score deltas under strongest blur, and the paired ACPC/PCC/CRA diagnostics move in the same direction. PushT and Cube do not show a clear score gain at the scale of training-seed and evaluation variance, and their diagnostic deltas are correspondingly weak or mixed. The selected rows follow the appendix convention: TwoRoom/Reacher blur are clear positive-transfer endpoints, while PushT/Cube resize are small-effect boundary endpoints. A full appendix audit over all task–family–seed endpoints for blur and resize gives the same directional readout: the composite ACPC/PCC/CRA/MAF rank has Spearman $\rho = 0.94$ with stress-success delta over 24 rows. We therefore use this table only as bounded scope evidence: ACPC-style paired diagnostics localize clear out-of-family improvements when present, but the current evidence does not justify a universal cross-perturbation robustness claim.

Table 12: Bounded unseen-stressor scope check. Score Δ is the fixed $\sigma_{\max} = 0.08$ checkpoint minus the no-noise baseline under the strongest unseen stress, averaged over training seeds 3072/3073/3074. Diagnostic deltas are from the matched paired-diagnostic slice averaged over training seeds 3072/3073/3074; negative is better for ACPC-H/transition and PCC, positive is better for CRA. The table is a scope check, not a universal transfer claim.

Task	stress	score Δ	Δ ACPC	Δ PCC	Δ CRA	reading
TwoRoom	blur $k = 15$	$+43.11 \pm 8.90$	-0.691	-44.7	$+0.616$	clear gain; diagnostics aligned
Reacher	blur $k = 15$	$+49.22 \pm 3.29$	-1.895	-48.9	$+0.587$	clear gain; diagnostics aligned
PushT	resize 0.25	$+2.89 \pm 17.98$	-0.267	-6.7	$+0.010$	no clear gain at this scale
Cube	resize 0.25	-0.89 ± 1.40	$+0.052$	-0.6	-0.030	no clear gain at this scale

6 Discussion and limitations

Scope. This paper is a controlled diagnostic study with four primary evidence layers: seed-level Gaussian behavior, three-seed fixed-rule diagnostic validation, margin-conditioned fixed-pool action-flip audit, and local task-feature proxy margin pass-rate. The three-training-seed Gaussian replication (canonical seed 3072 plus independent seeds 3073/3074) is the primary behavior statistic; evaluation-seed variance is kept as the within-seed measurement scale. Gaussian pixel noise remains the training/evaluation axis. Blur/resize scores, PLDM replication, and failure-case ablations are scope checks rather than additional main claims. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

Interpretation. The data challenge a too-strong reading of latent prediction: predicting future latents does not by itself guarantee robustness to visual noise. Encoder invariance is useful but incomplete. The sampled-pool bound identifies ACPC tails and clean margin tails as the diagnostic failure modes for one candidate pool, while the Gaussian linearization identifies the local object being measured as $J_{G_{\mathbf{a}}} J_E$. The useful target is agreement of action-conditioned predictions in task-relevant coordinates while retaining distinctions that affect actions, costs, or transitions. This reading is compatible with the task variation: TwoRoom can tolerate substantial nuisance contraction, PushT contact control needs fine pose resolution, Reacher sits in a lower-dimensional continuous regime, and Cube is less sensitive to global Gaussian noise in this sweep.

Using the diagnostics. The diagnostics are used as a ranked evidence chain, not as a single scalar. Closed-loop evaluation establishes the behavior; R_F and the rollout/cost readouts localize the predictive-dynamics change; the margin-conditioned flip audit calibrates the clean-margin side of the sampled-pool link on fixed candidate sets; the seed-3072-fixed rule audits whether the same readouts stay near held-out plateaus without replacing closed-loop evaluation; and the local task-feature proxy margin checks whether contraction preserves coarse action-relevant distinctions. Cube remains the useful boundary case, so point-best $\sigma_{\max}$ values are interpreted as plateau context rather than as a fine-grained ranking target.

Failure cases and method path. Two failure cases bound the method story without taking over the main contribution. The heteroscedastic NLL probe collapses PushT because contact-control transitions are high-error and action-critical; prediction difficulty alone is the wrong gate. The target-view ablation in Appendix N rules out perturbed-history $\rightarrow$ original-future targets as a sufficient closed-loop fix. The diagnostic route is now operational but deliberately modest: the seed-3072-fixed rule provides a plateau-localization audit on the full Gaussian grid, the margin-conditioned flip audit checks the fixed-pool clean-margin term, and the local task-feature proxy margin pass-rate checks selective consistency without claiming oracle semantic coverage. A trained paired predictive-dynamics consistency objective remains the separate method-paper path after diagnostic validity is established.

From diagnostics to more generalizable world models. The Gaussian-noise study should be read as a diagnostic probe, not as an endpoint. It points to two foundation capabilities for more generalizable latent world models. First, the encoder should filter visual nuisance variation while preserving geometry, contact, goal relations, and other action-relevant state differences needed for planning. Second, the predictor should make these representations useful under intervention: clean and perturbed views of the same state should produce consistent action-conditioned rollouts and cost-relevant predictions under the same action sequence, while states or actions that imply different futures remain separable.

These requirements are broader than additive Gaussian noise. They are prerequisites for world models that can adapt predictions and plans under new appearances, goals, or dynamics from context rather than memorizing a fixed training distribution. This paper does not test such context adaptation directly; it provides diagnostics that expose where the current encoder–predictor–planner chain succeeds, where it collapses nuisance variation safely, and where stronger objectives must protect action-relevant distinctions.

7 Conclusion

This paper studies Gaussian-noise robustness in JEPA latent world-model control as a multi-stage diagnostic problem. Encoder geometry shows how visual perturbations enter the latent state, action-conditioned rollout consistency shows whether they affect predicted futures and candidate costs, and a local task-feature proxy margin checks whether robustness is bought by erasing coarse action-relevant distinctions. Controlled additive-noise tests show that latent prediction alone does not guarantee closed-loop robustness. Input-side noise augmentation supplies recovered checkpoints for analyzing this failure mode, and the closed-loop recovery replicates across three training seeds.

The paired ACPC basin diagnostic localizes the recovery in rollout space; the seed-3072-fixed ACPC/PC-C/CRA/MAF rule gives a no-retraining plateau-localization audit on held-out training seeds; and the local task-feature proxy pass-rate checks that the recovered endpoint preserves local proxy-different rollout separation while shrinking same-state noisy radii. The theory is deliberately diagnostic: fixed-candidate ACPC bounds cost drift, sampled-pool ACPC exposes ACPC-tail and clean-margin failure modes, and local Gaussian linearization connects the measured basin radius to action-conditioned encoder–predictor sensitivity. The target-view ablation in Appendix N rules out perturbed-history $\rightarrow$ original-future targets as a sufficient closed-loop fix and leaves full-sequence perturbed targets as the empirical branch for this study. The broader implication is diagnostic rather than solved generalization: more generalizable world models need control-relevant representations and action-conditioned predictive dynamics that stay stable under nuisance variation without collapsing distinctions needed for planning. A paired predictive-dynamics training objective is the separate method-paper path suggested by these diagnostics.

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A Proofs and calibration for ACPC diagnostics

Proof of Theorem 1. Let $\mathcal{E}$ be the event that every sampled candidate satisfies $D_j \le \epsilon$. By the single-candidate tail assumption and a union bound,

$$\Pr_{\mathcal{A}}[\mathcal{E}^c] \le \sum_{j=1}^{K} \Pr[D_j > \epsilon] \le K\delta.$$

On $\mathcal{E}$, Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \le L_J\epsilon$ for all candidates in the sampled pool. If also $\Delta_{\mathcal{A}} > 2L_J\epsilon$, Proposition 3 implies that the clean top-1 candidate remains top-1 on the perturbed branch. Therefore a flip can occur only if either $\mathcal{E}^c$ occurs or the sampled clean margin is at most $2L_J\epsilon$:

$$\Pr[\text{flip}] \le \Pr[\mathcal{E}^c] + \Pr[\Delta_{\mathcal{A}} \le 2L_J\epsilon] \le K\delta + \Pr[\Delta_{\mathcal{A}} \le 2L_J\epsilon].$$

Proof of Corollary 2. Let F_t be the event that the clean and perturbed branches select different top-1 candidates at replanning step t for the sampled pool at that step. The sampled-pool theorem gives $\Pr[F_t] \le r_t$ under the stated per-step assumptions. The probability of at least one flip is bounded by another union bound:

$$\Pr[\exists t \le T : F_t] \le \sum_{t=1}^{T} \Pr[F_t] \le \sum_{t=1}^{T} r_t.$$

No independence across replanning steps is required for this bound. It is intentionally loose and does not account for environment feedback after a flip.

Proof of Proposition 4. For a fixed threshold ϵ, the variables $X_i = \mathbf{1}[D_i > \epsilon]$ are independent Bernoulli variables under the diagnostic sampling distribution. The one-sided Hoeffding inequality gives

$$\Pr[p_\epsilon > \hat{p}_\epsilon + u] \le \exp(-2nu^2).$$

Setting $u = \sqrt{\log(1/\delta)/(2n)}$ gives the stated upper confidence bound. The same argument applies to any bounded Bernoulli diagnostic, including a margin-conditioned action-flip indicator, after fixing the diagnostic sampling distribution and candidate construction, with the same independence caveat.

Derivation of Proposition 5. By Taylor expansion of the encoder around o,

$$E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + R_E(\xi), \qquad \|R_E(\xi)\| = O(\|\xi\|^2).$$

Expanding $G_{\mathbf{a}}$ around $E_\theta(o)$ and substituting the encoder expansion gives

$$G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

with $\|R_\xi\| = O(\|\xi\|^2)$ under the bounded second-order remainder assumptions. For $\xi \sim \mathcal{N}(0, \sigma^2 I)$ and $A = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)$,

$$\mathbb{E}\|A\xi\|_2^2 = \operatorname{tr}(A\,\mathbb{E}[\xi\xi^\top]A^\top) = \sigma^2\|A\|_F^2.$$

The cross and remainder terms are higher order because $\mathbb{E}\|\xi\|^3 = O(\sigma^3)$ and $\mathbb{E}\|\xi\|^4 = O(\sigma^4)$, yielding the stated expansion. This is a local small-noise calibration, not a finite-noise robustness theorem.

Hoeffding calibration for the local task-feature proxy margin. Let $M_i \in \{0, 1\}$ denote the pass event $M_i = \mathbf{1}[d_{\text{proxy diff},i} > d_{\text{same state noise},i} + \delta_m]$ for sampled pair i, and let $\hat{p} = n^{-1} \sum_i M_i$. Under independent-pair calibration, Hoeffding's inequality gives

$$\Pr[p < \hat{p} - \epsilon] \le \exp(-2n\epsilon^2),$$

so with probability at least $1 - \alpha$,

$$p \ge \hat{p} - \sqrt{\frac{\log(1/\alpha)}{2n}}.$$

For the reported high-noise endpoints, each task/endpoint has $n = 300$ pairs after aggregating 100 pairs over three training seeds. A pass-rate $\hat{p} = 1.00$ gives $1 - \sqrt{\log 20/600} = 0.929$ at 95% confidence. Because adjacent dataset windows can be dependent, the main text uses this as calibration rather than as a dataset-wide guarantee.

One-step prediction-loss triangle. For one-step predictions, write $\hat{z}_{t+1} = F_\theta(E_\theta(h_t), a_t)$, $\hat{\tilde{z}}_{t+1} = F_\theta(E_\theta(\tilde{h}_t), a_t)$, $z_{t+1} = E_\theta(h_{t+1})$, and $\tilde{z}_{t+1} = E_\theta(\tilde{h}_{t+1})$. Any metric d satisfies

$$d(\hat{z}_{t+1}, \hat{\tilde{z}}_{t+1}) \le d(\hat{z}_{t+1}, z_{t+1}) + d(z_{t+1}, \tilde{z}_{t+1}) + d(\tilde{z}_{t+1}, \hat{\tilde{z}}_{t+1}).$$

Thus clean prediction error, noisy-branch prediction error, and target-future nuisance stability jointly upper-bound one-step predictive disagreement. This teacher-forced one-step inequality does not guarantee rollout stability or closed-loop robustness; the target-view ablation in Appendix N shows that perturbed-history to original-future denoising is not sufficient in this study.

B Related-work boundary table

Table 13: Boundary to neighboring consistency and control-representation work.

Neighboring thread	Shared concern	Boundary in this paper
ViGMO [28]	Visual distractions and latent consistency	We do not propose a generic encoder-consistency regularizer; our diagnostic places consistency after action-conditioned rollout and adds a discriminability guard.
ReOI [6]	Robust visual MPC under distractors	We do not propose an observation-intervention policy; we audit existing JEPA checkpoints with paired same-action clean/noised rollouts and fixed-candidate stability conditions.
Bisimulation / Bisim-JEPA [11, 44, 38]	Control-relevant state abstraction	Same-state perturbed and unperturbed views need not share an encoder latent; agreement is required at the predictive-dynamics level.
LeJEPA theory [19]	Latent variables and latent planning	We do not claim identifiability; we test whether learned checkpoints preserve predictive consistency under visual perturbation.
Group-action world models [42]	Action-faithful rollout structure	We test same-action paired perturbation consistency, not identity/inverse/composition consistency across action sequences.
Value-aware models [13, 40]	Models should serve downstream control	We make this operational as paired action-conditioned rollout agreement, not value equivalence or value prediction.
V-JEPA family [4, 3, 17]	Latent video prediction and noisy-environment probes	Representation recovery and closed-loop control stress different properties; our probe measures eval-time closed-loop behavior.

C Experimental details

C.1 Environments

- **PushT.** 2D continuous pushing; 20,000 expert episodes; $\sim$196 steps/episode; action dim 2 (direction + magnitude); 224×224 RGB.

- **TwoRoom.** 2D continuous navigation; 10,000 episodes; ~ 92 steps; action dim 2; 224×224 RGB.
- **Reacher.** 2D arm control (DeepMind Control Suite); 10,000 episodes; 200 steps; action dim 2.
- **Cube.** 3D cube manipulation (OGBench); 10,000 episodes; 200 steps; action dim 7.

C.2 Noise augmentation implementation

The actual implementation is `utils.py::AddNormalizedGaussianNoise`. Key points: (i) noise is added on ImageNet-normalized tensors and must be divided by the per-channel std to retain "pixel-space std" semantics; (ii) sampling is per-frame independent over the leading frame dims, not per-batch; (iii) all sweeps in this paper fix the noise probability to 1.0, set the lower std to 0, and vary only the upper std $\sigma_{\max}$.

```
class AddNormalizedGaussianNoise:
    """Per-frame independent. Each frame draws Bernoulli(noise_prob) then
    std ~ Uniform(std_min, std_max). Pixel-space std is converted to
    normalized space by dividing by per-channel std before adding."""
    def __init__(self, std_min, std_max, noise_prob=1.0):
        self.std_low, self.std_high, self.noise_prob = std_min, std_max, noise_prob
        self.channel_std = torch.as_tensor(IMAGENET_STD) # (C,)

    def __call__(self, x):
        if self.std_high <= 0 or self.noise_prob <= 0:
            return x
        leading = x.shape[:-3]
        stds = torch.empty(leading, device=x.device, dtype=x.dtype).uniform_(
            self.std_low, self.std_high)
        if self.noise_prob < 1.0:
            mask = (torch.rand(leading, device=x.device) < self.noise_prob).to(x.dtype)
            stds = stds * mask
        per_frame_scale = stds.view(*leading, 1, 1, 1)
        channel_factor = (1.0 / self.channel_std.to(x.device, x.dtype)).view(
            *([1] * len(leading)), -1, 1, 1)
        scale = per_frame_scale * channel_factor # pixel-space std -> normalized
        return x + torch.randn_like(x) * scale
```

C.3 Evaluation protocol

Unperturbed evaluation uses no noise, 100 trajectories per seed × 3 seeds (42/43/44), totalling 300 trajectories per condition. The primary noisy evaluation adds Gaussian noise of std $\in \{0.05, 0.08\}$ to observation pixels while leaving the goal image unperturbed under the same 3-seed protocol. Observation+goal Gaussian noise is retained as an auxiliary full-input-stream stress condition and is reported separately where used. Success criterion is task-specific (PushT: T-block pose match; TwoRoom: target region; Reacher: joint-angle match; Cube: cube position match).

C.4 Full LeWM noise-sweep tables

Tables 14 and 15 give the complete LeWM sweep behind Table 3 and fig. 2. All cells are success rate mean ± population std across the three evaluation seeds.

Table 14: LeWM+noise sweep, unperturbed success rate (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	94.00 ± 3.56	86.33 ± 2.36	58.67 ± 1.25	66.67 ± 2.62
0.01	93.67 ± 3.30	88.00 ± 3.74	61.67 ± 2.49	69.33 ± 0.47
0.02	95.00 ± 2.83	88.33 ± 2.62	85.67 ± 2.49	60.00 ± 1.63
0.03	96.33 ± 3.30	89.67 ± 1.70	78.67 ± 1.25	65.00 ± 1.63
0.04	96.33 ± 2.05	89.33 ± 2.05	84.00 ± 2.94	69.00 ± 3.74
0.05	96.00 ± 2.83	80.67 ± 4.78	70.00 ± 2.16	59.33 ± 0.94
0.06	96.67 ± 2.05	89.33 ± 2.05	86.00 ± 2.94	66.67 ± 2.05
0.07	96.00 ± 1.63	85.67 ± 3.09	83.67 ± 3.30	67.67 ± 0.94
0.08	98.33 ± 0.47	88.33 ± 2.87	84.00 ± 0.82	62.33 ± 1.25

Table 15: LeWM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%).

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	65.67 ± 1.25	4.33 ± 1.25	18.33 ± 2.05	47.00 ± 6.38
0.01	94.00 ± 2.83	51.67 ± 5.44	45.67 ± 3.86	49.00 ± 2.45
0.02	94.33 ± 3.09	73.67 ± 4.99	83.67 ± 2.49	58.67 ± 3.30
0.03	97.00 ± 2.94	82.00 ± 6.48	78.67 ± 2.87	68.33 ± 2.49
0.04	96.67 ± 2.05	86.67 ± 2.87	82.00 ± 0.82	65.00 ± 2.45
0.05	95.33 ± 3.30	78.33 ± 5.91	70.00 ± 2.94	60.00 ± 0.82
0.06	96.33 ± 2.05	87.67 ± 3.30	81.33 ± 1.70	66.67 ± 1.70
0.07	96.67 ± 1.25	84.00 ± 3.74	84.67 ± 5.19	67.33 ± 2.05
0.08	97.67 ± 0.94	89.00 ± 4.32	82.00 ± 1.41	62.00 ± 1.41

C.5 Full LeWM ACPC-basin grid

Table 16: Full LeWM Gaussian-noise ACPC-basin grid. Success is observation-noise 0.08 success with an unperturbed goal; drop is unperturbed success minus that success. Negative drop means no degradation within this evaluation estimate. R_E and R_F are the normalized encoder-history and 8-step rollout-view radii used in Table 7.

Task	$\sigma_{\max}$	unpert.	obs 0.08	drop	R_E	R_F	R_F/R_E	note
TwoRoom	0	94.00	65.67	28.33	3.538	0.785	0.222	base
TwoRoom	0.01	93.67	94.00	-0.33	1.633	0.232	0.142	
TwoRoom	0.02	95.00	94.33	0.67	0.931	0.124	0.133	
TwoRoom	0.03	96.33	97.00	-0.67	0.720	0.084	0.117	
TwoRoom	0.04	96.33	96.67	-0.33	0.518	0.056	0.108	
TwoRoom	0.05	96.00	95.33	0.67	0.360	0.039	0.108	
TwoRoom	0.06	96.67	96.33	0.33	0.300	0.034	0.112	
TwoRoom	0.07	96.00	96.67	-0.67	0.259	0.029	0.112	
TwoRoom	0.08	98.33	97.67	0.67	0.235	0.028	0.121	obs0.08 ref
PushT	0	86.33	4.33	82.00	1.727	1.543	0.893	base
PushT	0.01	88.00	51.67	36.33	0.677	0.474	0.701	
PushT	0.02	88.33	73.67	14.67	0.383	0.250	0.654	
PushT	0.03	89.67	82.00	7.67	0.253	0.177	0.699	
PushT	0.04	89.33	86.67	2.67	0.222	0.145	0.653	
PushT	0.05	80.67	78.33	2.33	0.232	0.141	0.607	
PushT	0.06	89.33	87.67	1.67	0.169	0.106	0.628	
PushT	0.07	85.67	84.00	1.67	0.159	0.105	0.663	
PushT	0.08	88.33	89.00	-0.67	0.137	0.088	0.642	obs0.08 ref
Reacher	0	58.67	18.33	40.33	4.238	1.012	0.239	base
Reacher	0.01	61.67	45.67	16.00	1.359	0.241	0.177	
Reacher	0.02	85.67	83.67	2.00	0.067	0.017	0.251	
Reacher	0.03	78.67	78.67	0.00	0.283	0.052	0.185	
Reacher	0.04	84.00	82.00	2.00	0.200	0.038	0.191	
Reacher	0.05	70.00	70.00	0.00	0.064	0.015	0.241	
Reacher	0.06	86.00	81.33	4.67	0.147	0.031	0.209	
Reacher	0.07	83.67	84.67	-1.00	0.135	0.027	0.200	obs0.08 ref
Reacher	0.08	84.00	82.00	2.00	0.116	0.024	0.209	
Cube	0	66.67	47.00	19.67	1.343	0.957	0.713	base
Cube	0.01	69.33	49.00	20.33	0.772	0.471	0.610	
Cube	0.02	60.00	58.67	1.33	0.063	0.019	0.293	
Cube	0.03	65.00	68.33	-3.33	0.238	0.115	0.481	obs0.08 ref
Cube	0.04	69.00	65.00	4.00	0.177	0.078	0.440	
Cube	0.05	59.33	60.00	-0.67	0.045	0.010	0.232	
Cube	0.06	66.67	66.67	0.00	0.120	0.046	0.385	
Cube	0.07	67.67	67.33	0.33	0.098	0.038	0.390	
Cube	0.08	62.33	62.00	0.33	0.094	0.036	0.385	

C.6 Compute

Training runs on a single NVIDIA H800 (80 GB) and follows the LeWM training schedule released with the baseline.

C.7 Main-figure rendering

Figure 2 is produced by the main plot script listed in Appendix C.10; no checkpoint or model loading is required.

Figure 1 is produced by `tools/paper1_selective_contraction.py`. The rendering command used for this paper is:

```
python -m tools.paper1_selective_contraction \
  --plot-clusters --plot-tasks PushT \
  --n-sequences 128 --cluster-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --cluster-perturb-repeats 6 \
  --feature-cache-dir /tmp/paper1_selective_contraction_cache \
  --cluster-out-dir assets/paper1_figs \
  --cluster-envelope ellipse --cluster-envelope-coverage 0.90
```

Unlike the aggregate figure generator, this optional rendering path loads the PushT checkpoints and dataset windows to draw repeated same-state perturbation samples. The feature-cache directory stores the extracted encoder and rollout feature arrays as parameter-keyed `.npz` files, so label/style-only redraws do not reload checkpoints. Use `--refresh-feature-cache` only when changing checkpoints, sampled windows, view stds, or feature-extraction parameters. The small panel summaries report high-dimensional statistics; the t-SNE ellipses are only qualitative 2-D covariance envelopes.

C.8 Local cluster atlas: a projection-free check of the basin visualization

Because t-SNE distorts global distances, Figure 3 provides a complementary *local normalized atlas* view of the same PushT data. Each small box selects one original dataset state; the circle radius equals the distance to its nearest *other* original state, computed in the original high-dimensional feature space, and the color markers are that state's original and Gaussian-perturbed views after a local PCA used only for in-box display. The visual question per box is therefore exactly the high-dimensional one: does the same-state perturbation cloud stay inside the nearest-different-state radius? On the no-noise-trained checkpoint many perturbation views escape their circles (median $r/\mathrm{NN} > 1$, few disjoint balls); on the full-sequence noise-trained checkpoint the clouds contract well inside the circles in both encoder features and the ACPC rollout readout R_F. This avoids the global-projection caveats of Figure 1 at the cost of showing only local neighborhoods; the quantitative multi-task evidence remains Table 7.

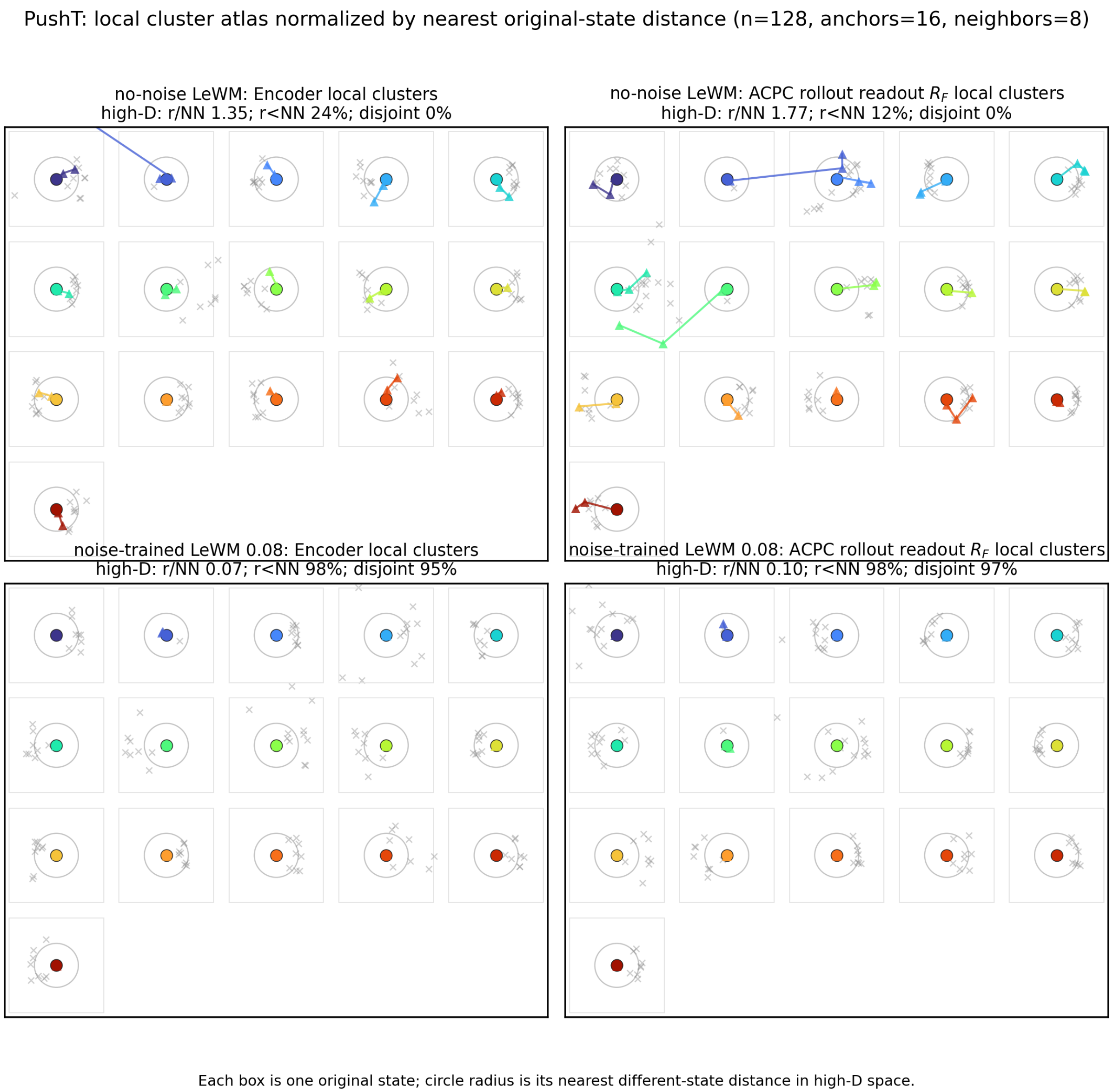


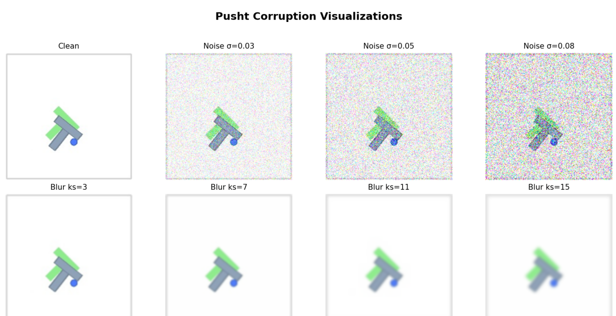
Figure 3: PushT local cluster atlas ($n = 128$ windows, 16 anchors, 8 neighbors). Each box is one original state with Gaussian-perturbed views after local PCA; the circle radius is the high-dimensional nearest-other-state distance. Panel summaries report original-space local statistics.

The atlas rendering command is:

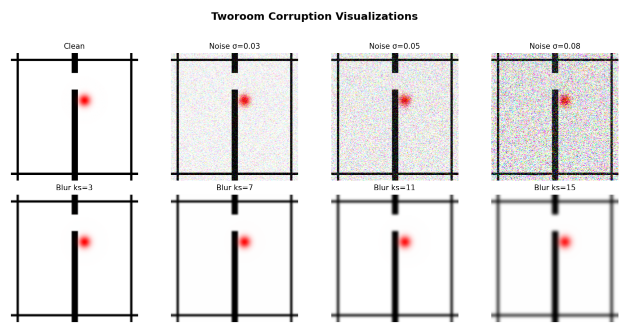
```
python -m tools.paper1_selective_contraction \
  --plot-atlas --plot-tasks PushT \
  --n-sequences 128 --atlas-anchor-count 16 \
  --view-stds 0.0 0.01 0.04 0.08 \
  --feature-cache-dir /tmp/paper1_selective_contraction_cache \
  --atlas-out-dir assets/paper1_figs
```

C.9 Perturbation visualizations

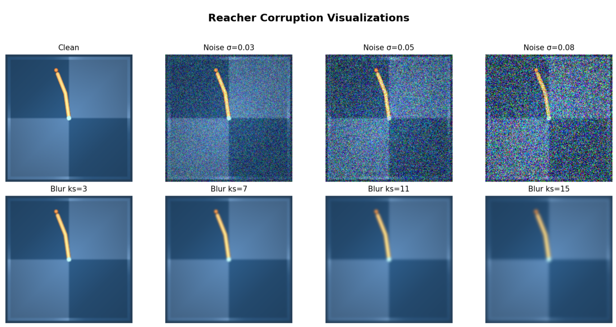
Figure 4 shows representative renderings of the two evaluated perturbation families, applied to a sample observation from each of the four tasks. The Gaussian-noise row includes a mild calibration severity $\sigma = 0.03$ in addition to the two evaluation endpoints $\sigma \in \{0.05, 0.08\}$ used in Sections 5.1 and 5.2; the main sweep treats observation-only noise with an unperturbed goal as the primary endpoint and keeps observation+goal noise as an auxiliary stress condition. The Gaussian-blur kernel sizes ($k \in \{3, 7, 11, 15\}$) match the stress test in Appendix J. Each panel is a single 224×224 RGB frame from the corresponding task's evaluation rollout, after perturbation and before encoder normalization.



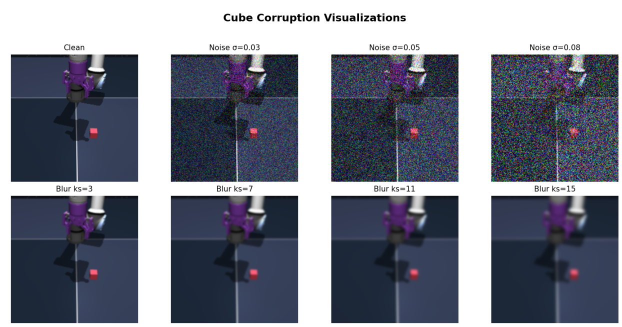
(a) PushT



(b) TwoRoom



(c) Reacher



(d) Cube

Figure 4: Perturbation visualizations on the four tasks. Each panel is a 2×4 grid: additive Gaussian pixel noise on row 1 (Original, $\sigma \in \{0.03, 0.05, 0.08\}$) and Gaussian blur on row 2 ($k \in \{3, 7, 11, 15\}$, auto sigma). The $\sigma = 0.08$ endpoint is the primary Gaussian-noise stress condition used in Tables 2 and 15.

C.10 Artifact and key-name mapping

The main text uses reader-facing diagnostic names. Exact relative artifact paths and implementation keys are collected here so that the manuscript remains readable while preserving reproducibility.

Table 17: Mapping from paper names to released relative paths and implementation keys.

Paper name	Released relative path or key
training-noise ceiling $\sigma_{\max}$	`std_max`
canonical evaluation artifact	`assets/paper1_data/canonical_evals_20260517.json`
canonical diagnostic artifact	`assets/paper1_data/canonical_diagnostics_20260517.json`
LeWM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics.json`
No-retraining diagnostic audit	`assets/paper1_data/no_retrain_diagnostic_audit.json`
selector-baseline audit	`assets/paper1_data/selector_baseline_audit_20260704.json`
margin-conditioned flip artifact	`assets/paper1_data/margin_flip_curve_lewm_three_seed.json`
local task-feature margin artifact	`assets/paper1_data/semantic_local_margin_lewm_three_seed.json`
training-seed Gaussian lockbox artifact	`assets/paper1_data/training_seed_gaussian_lockbox.json`
validation remediation summary	`assets/paper1_data/prospective_validation_summary.json`
PLDM ACPC-basin artifact	`assets/paper1_data/acpc_basin_diagnostics_pldm.json`
Phase-0 paired ACPC artifact	`assets/paper1_data/acpc_phase0_clean_goal_seed9101.json`
unseen Phase-0 ACPC subset artifact	`assets/paper1_data/unseen_phase0_acpc_subset.json`
blur baseline artifact	`assets/paper1_data/canonical_blur_baselines_20260523.json`
target-view ablation artifact	`assets/paper1_data/target_view_closed_loop_summary.json`
PLDM full-diagnostic artifact	`assets/paper1_data/canonical_full_diagnostics_pldm_20260523.json`
encoder-shift ratio	`noise_to_nn_cos_ratio`
fragility ratio	`predictor_target_to_nn_cos_ratio_at_max_std`
transition-resolution ratio	`transition_resolution_ratio_{cos,l2}`
8-step rollout drift	`predictor_rollout_T8_l2`
ID probe R^2	`id_probe_r2`
main plot script	`tools/paper1_figs.py`
ACPC basin script	`tools/paper1_acpc_basin.py`
No-retraining audit script	`tools/paper1_no_retrain_remediation.py`
training-seed lockbox script	`tools/paper1_training_seed_lockbox.py`
validation remediation script	`tools/paper1_validation_remediation.py`
selective-contraction rendering script	`tools/paper1_selective_contraction.py`

D Diagnostic framework definitions

D.1 LeWM training objective

LeWM [26] is trained with latent prediction and SIGReg regularization:

$$\mathcal{L}_{\text{LeWM}} = \mathcal{L}_{\text{pred}} + \lambda \cdot \mathcal{L}_{\text{SIGReg}}, \tag{9}$$

where $\mathcal{L}_{\text{pred}}$ is the latent-space MSE between predicted and target representations. SIGReg projects each latent onto unit-norm random directions, computes the Epps–Pulley empirical-characteristic-function distance [8] between each projection and $\mathcal{N}(0,1)$, and aggregates with Gauss-window weights to prevent collapse without explicit BatchNorm. Inference uses CEM [22] for latent-space MPC [9].

D.2 Five-layer diagnostic framework

The diagnostic framework decomposes the JEPA world-model control pass into pixels $\to$ encoder $f \to$ latent $z \to$ predictor $g \to$ cost surface $\to$ planned actions. Layers 1–2 describe whether visual perturbations enter the latent and how they change latent geometry. Layer 3 measures how the predictor amplifies an encoder shift. Layer 4 injects noise directly into z to separate encoder effects from predictor and cost effects. Layer 5 checks task-relevant information retained in the latent. We use nearest-neighbor distance as a local latent scale primitive in the spirit of kNN out-of-distribution detection [35]; the scale-normalized ratios are introduced for this diagnostic study.

Minimal notation. Let $z_i = f(x_i)$ and $\tilde{z}_i = f(\tilde{x}_i)$ be unperturbed and noised latents for the same token. The distance, local nearest-neighbor (NN) scale, and three introduced ratios are

$$\begin{aligned} d_{\cos}(a,b) &= 1 - \frac{a^\top b}{\|a\|\,\|b\|}, \\ d_i^{\mathrm{NN}} &= \min_{j \neq i} d_{\cos}(z_i, z_j), \\ r_i^{\mathrm{enc}} &= \frac{d_{\cos}(z_i, \tilde{z}_i)}{d_i^{\mathrm{NN}} + \epsilon}, \\ r_i^{\mathrm{pred}} &= \frac{d_{\cos}\big(g(z)_i, g(\tilde{z})_i\big)}{d_i^{\mathrm{NN}} + \epsilon}, \\ R_d &= \frac{\mathrm{median}_t \ d(z_t, z_{t+1})}{\mathrm{median}_{(t,t') \in \mathcal{F}} \ d(z_t, z_{t'}) + \epsilon}. \end{aligned} \tag{10}$$

Here r_i^{enc} is the encoder-shift ratio, r_i^{pred} is the fragility ratio, and R_d is the transition-resolution ratio for either cosine or L2 distance d. $\mathcal{F}$ denotes random far pairs from different temporal locations, and $\epsilon = 10^{-8}$ is a numerical-stability constant.

Layer metrics. Layer 1 reports raw angular/L2 encoder shift, angular gain per unit noise, and the encoder-shift ratio in Equation (10). Layer 2 reports local neighborhood scale, effective rank [31], and CKA [20]. Layer 3 reports fragility ratio and multi-step rollout L2 drift. Layer 4 reports latent-noise cost-surface slope and latent robust radius. Layer 5 reports the L2/cosine transition-resolution ratio and inverse-dynamics linear-probe R^2 [1]. All five layers are descriptive diagnostics used to localize what changed at evaluated checkpoints; none is treated as a standalone checkpoint selector.

E Full diagnostic profile of LeWM-base on four tasks

Table 18 summarizes every core diagnostic on the four LeWM-base checkpoints. Exact released keys and diagnostic families are mapped in Appendix C.10; the table itself uses reader-facing metric names.

Table 18: Full LeWM-base diagnostics across four tasks.

Layer / metric	TwoRoom	PushT	Reacher	Cube	Unit / note
Encoder geometry					
Local NN cosine distance	0.0449	0.2360	0.0633	0.1856	cosine distance
Pairwise cosine distance	0.9904	1.0228	1.0252	1.0193	pair-wise cos dist
Effective rank	47.60	76.42	61.04	73.25	effective rank
Noise sensitivity					
Median noise angle (@ std 0.05)	5.51°	1.33°	3.22°	1.40°	median angular shift
Encoder-shift ratio	0.1031	0.011	0.0249	0.016	noise/NN cos ratio
Robust radius	0.0142	0.0537	0.0142	0.0356	critical noise std
First high-risk std	>0.08	>0.08	>0.08	>0.08	first high-risk std
Angular gain per std	1085.8	284.8	831.7	327.0	°/std angular gain
Geometry label	balanced	robust	balanced	robust	geometry label
Task resolution					
transition-resolution ratio (L2)	0.7216	0.3015	0.3704	0.4847	L2 resolution ratio
transition-resolution ratio (cos)	0.5538	0.0868	0.1351	0.2347	cos resolution ratio
ID probe R^2	0.2889	0.7739	0.1621	0.6657	linear probe R^2
Minimum ID probe R^2	0.2599	0.6786	0.1366	0.0972	min probe R^2
LiDAR rank proxy	46.06	13.95	45.90	42.46	LiDAR rank proxy
Action effect					
Mean action-induced prediction shift	0.5329	0.1283	0.2518	0.2364	action-perturb mean shift
Action-shift correlation	0.2847	0.2873	0.4042	0.2559	shift $\times$ $\|a\|$ corr
Predictor stability					
target/NN ratio	1.51e-4	3.54e-6	2.67e-5	3.39e-6	max-std target/NN ratio
Multi-step rollout drift	18.62	18.65	15.17	20.20	T=8 rollout L2 drift
Latent noise					
Linear CKA at max std	0.1986	0.5536	0.3085	0.1814	CKA unperturbed vs noisy
Latent cost-surface slope	635.31	1.3886	599.45	0.6208	goal latent cost slope

F Heteroscedastic-loss formulation

The scale-preserving heteroscedastic NLL referenced in Section 6 and detailed in Appendix G is:

$$\mathcal{L}_{\text{hetero}} = \tfrac{1}{2}\,\exp(-s_t)\,\|z_{t+1} - \hat{z}_{t+1}\|^2 + \tfrac{1}{2}\,s_t\,, \tag{11}$$

where s_t is the σ-head-predicted log-variance. During training $\exp(-s_t)$ acts as an automatic weight: high-error transitions are down-weighted and low-error transitions are up-weighted. When $s_t \equiv 0$ the loss reduces to plain MSE.

G Heteroscedastic-loss negative result (data)

This appendix records the full heteroscedastic-loss data referenced in Section 6. The σ-head is trained jointly with the predictor; gradient is detached on the σ path so the mean prediction path is exactly LeWM MSE when σ is constant.

Table 19: Heteroscedastic-loss evaluation (canonical 3-seed $\times$ 100 protocol). The LeWM+noise rows use representative observation+goal 0.08 high-noise rows (TwoRoom $\sigma_{\max} = 0.08$, PushT $\sigma_{\max} = 0.06$), matching the strongest stress column of this ablation; the observation-only representative rows in Table 15 differ.

Task / model	unpert.	goal .05	obs .05	obs+goal .05	goal .08	obs+goal .08
TwoRoom LeWM-base	94.00	73.33	72.33	61.33	58.67	50.00
TwoRoom LeWM+noise row	98.33	98.00	98.33	98.00	98.67	98.67
TwoRoom hetero	99.67	85.33	96.67	84.67	73.33	55.33
PushT LeWM-base	86.33	38.00	17.00	12.00	11.33	4.67
PushT LeWM+noise row	89.33	87.67	88.00	88.33	89.67	87.00
PushT hetero	13.33	7.67	7.67	7.67	9.67	6.00

TwoRoom hetero reaches 99.67% on unperturbed evaluation — compatible with the prior that low-dimensional discrete tasks can tolerate stronger nuisance contraction / clustering — but lags noise training on high-noise robustness. **PushT hetero unperturbed success is** 13.33% — **a method-level failure**, not evidence for a useful robustness compromise.

Table 20: Representation diagnostics of the hetero-loss failure.

Metric	TwoRoom base	TwoRoom hetero	PushT base	PushT hetero
Local NN cosine distance	0.0449	0.0281	0.2360	0.1051
Effective rank	47.60	33.59	76.42	42.85
transition-resolution ratio (cos)	0.5538	0.3780	0.0868	0.0101
transition-resolution ratio (L2)	0.7216	0.6055	0.3015	0.1023
ID probe R^2	0.2889	−0.0573	0.7739	0.2678
Mean action-induced prediction shift	0.5329	0.4482	0.1283	0.0841
Multi-step rollout drift	18.62	17.90	18.65	14.01

Hetero-loss compresses the representation in both tasks. In TwoRoom (low-dimensional, discrete, redundant), the observed compression coexists with high unperturbed success, but the robustness shortfall shows that compression alone is not the target. For PushT, the L2 transition-resolution ratio collapses from 0.3015 to 0.1023 and ID probe R^2 from 0.7739 to 0.2678 — task-relevant state information is erased. The contrast with input-side noise training is sharp: the fixed $\sigma_{\max} = 0.08$ PushT checkpoint in Table 10 leaves rank and probe nearly unchanged, so this collapse is specific to error-based downweighting. The drop in multi-step rollout drift is not good news either: the latent has become more *predictable* without being more *controllable*.

Mechanism. The learned σ values align with visible high-error transition regimes (contact frames in PushT, doorway crossings in TwoRoom, etc.), but using σ as an automatic weight to down-weight high-error transitions misclassifies the contact-and-control-critical states of PushT as "unimportant" and erases them. This is a negative example of the broader selective-consistency lesson: in contact-heavy control, **hard $\neq$ unimportant**.

This finding motivates a narrower follow-up direction: per-token *consistency* controllers should be separated from loss reweighting, and any difficulty signal should be detached from the mean prediction path unless it is explicitly protected by an action-relevance guard. We leave that objective design outside the present diagnostic study.

H Mechanism-attribution probe

Section 5.6 reports what moves in the representation. To localize whether the perturbation enters through the encoder or later predictor/planner stages, we use latent-noise probing as a conservative mechanism check. Directly injecting noise into latent z skips the encoder and decouples encoder contributions from predictor and cost contributions.

Table 21: Three-layer attribution by task (LeWM $n = 9$ sweep, unified 3-seed $\times$ 100 eval). These probes are descriptive mechanism readouts, not checkpoint selectors.

Task	Strongest signal in these probes	Bounded interpretation
TwoRoom	encoder-shift signal (rank-saturated)	saturation limits mechanism inference
PushT	encoder + single-step predictor	contact-relevant resolution must be protected
Reacher	encoder + multi-step rollout	rollout stabilization aligns with the representative diagnostic movement
Cube	encoder	weaker behavioral recovery despite rollout stabilization

Across these probes, the strongest common signal is encoder-side perturbation transduced by the predictor. We use this as mechanism localization only: the sweep has broad plateaus and finite evaluation variance, so these scalar readouts should not be used to rank checkpoints inside a plateau. A stronger causal claim about planner or cost-surface contributions would require multi-task latent-injection or cost-ranking ablations.

I Cross-method replication: PLDM noise sweep on four tasks

This appendix records the second method-family on which the Gaussian-noise sweep and ACPC-basin diagnostics replicate. PLDM follows the joint-embedding predictive line from Sobal et al. [33]; the concrete latent-dynamics planning baseline is from Sobal et al. [34]. We use the implementation distributed through the `stable-worldmodel` baseline suite [25]. Training and evaluation follow the LeWM canonical protocol bit-for-bit: per-frame Gaussian pixel noise, $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$, three evaluation seeds (42/43/44), 100 trajectories per seed.

Coverage. The PLDM release is complete for the same grid as the LeWM canonical sweep: 4 tasks $\times$ 9 configurations (no-noise baseline plus $\sigma_{\max} \in \{0.01, \ldots, 0.08\}$), three evaluation seeds (42/43/44), and 100 trajectories per seed. The PLDM eval, predictor-diagnostic, full-diagnostic, and ACPC-basin JSON files are listed in the data manifest; we do *not* merge them into the LeWM canonical files used by the main LeWM tables.

No-noise-trained PLDM cliff. PLDM reproduces the no-noise-trained Gaussian-noise cliff on PushT and TwoRoom, while Reacher and Cube show much smaller drops under this PLDM training recipe (Table 22). This supports a method-family reading of the failure mode without claiming every task has the same cliff magnitude.

Table 22: PLDM baseline trained without input-noise augmentation on all four tasks. Success rate mean $\pm$ population std, 3 evaluation seeds $\times$ 100 trajectories. Drop is clean success minus observation-noise 0.08 success.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	96.67 ± 1.70	79.67 ± 4.03	67.00 ± 2.16	51.67 ± 2.05	29.67
PushT	75.33 ± 3.68	46.00 ± 4.97	18.33 ± 2.05	10.00 ± 2.16	57.00
Reacher	82.67 ± 1.70	81.33 ± 4.03	80.33 ± 5.73	79.33 ± 0.94	2.33
Cube	52.67 ± 3.30	50.33 ± 4.50	48.33 ± 4.50	48.33 ± 2.87	4.33

PLDM full sweep — unperturbed evaluation. Table 23 reports the per-task unperturbed success rate at every trained $\sigma_{\max}$ level. Table 24 reports the corresponding observation-noise 0.08 robustness with an unperturbed goal.

Table 23: PLDM+noise sweep, unperturbed success rate (%). † marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	96.67 ± 1.70	75.33 ± 3.68	82.67 ± 1.70	52.67 ± 3.30
0.01	96.33 ± 0.94	72.33 ± 4.19	78.00 ± 0.82	51.33 ± 1.25
0.02	97.33 ± 0.47	71.33 ± 3.86	82.33 ± 2.36	53.33 ± 1.70
0.03	96.67 ± 1.70	$76.67 \pm 7.54^{\dagger}$	83.00 ± 3.74	52.67 ± 3.30
0.04	97.67 ± 0.47	68.67 ± 5.25	$85.00 \pm 2.16^{\dagger}$	54.00 ± 2.94
0.05	97.67 ± 1.25	74.33 ± 3.40	72.00 ± 1.41	54.00 ± 2.16
0.06	98.00 ± 0.82	70.33 ± 6.18	79.00 ± 0.82	$56.00 \pm 4.97^{\dagger}$
0.07	98.00 ± 0.82	66.67 ± 8.38	80.67 ± 2.62	53.67 ± 3.68
0.08	$98.33 \pm 0.94^{\dagger}$	68.00 ± 1.41	80.33 ± 1.25	54.00 ± 1.63

Table 24: PLDM+noise sweep, observation-noise 0.08 success rate with unperturbed goal (%). ‡ marks one representative high grid row per task for compact reading.

$\sigma_{\max}$	TwoRoom	PushT	Reacher	Cube
0 (base)	67.00 ± 2.16	18.33 ± 2.05	80.33 ± 5.73	48.33 ± 4.50
0.01	96.00 ± 1.63	23.33 ± 2.49	77.67 ± 3.40	51.33 ± 2.36
0.02	95.67 ± 0.94	47.00 ± 0.82	80.67 ± 5.25	51.00 ± 4.32
0.03	96.33 ± 1.25	$73.00 \pm 7.48^{\ddagger}$	$81.67 \pm 4.71^{\ddagger}$	54.33 ± 2.62
0.04	97.33 ± 0.94	66.67 ± 5.73	80.33 ± 2.87	$54.67 \pm 2.36^{\ddagger}$
0.05	97.67 ± 1.89	71.67 ± 5.79	62.67 ± 4.19	54.00 ± 3.74
0.06	$98.00 \pm 0.00^{\ddagger}$	69.33 ± 4.78	75.00 ± 0.82	54.33 ± 0.94
0.07	98.00 ± 0.82	64.00 ± 7.79	78.33 ± 2.05	53.00 ± 3.74
0.08	97.67 ± 1.25	68.00 ± 5.66	79.33 ± 1.25	53.67 ± 2.05

Reading. Four quantitative observations are useful for interpreting PLDM as a second method family:

1. **TwoRoom (visually redundant) shows large recovery on both methods.** PLDM observation-noise 0.08 success rises from 67.00% at the no-noise baseline to 98.00% at $\sigma_{\max} = 0.06$. This mirrors the LeWM reading that TwoRoom tolerates strong same-state nuisance contraction.
2. **PushT (contact-heavy) exhibits the Gaussian-noise cliff and large noise-training recovery.** PLDM without noise augmentation drops by 57.00 pt under observation-noise 0.08; training with $\sigma_{\max} = 0.03$ recovers observation-noise 0.08 success to 73.00% (+54.67 pt over the no-noise baseline). The unperturbed and observation-noise 0.08 representative high rows are co-located at $\sigma_{\max} = 0.03$ on PLDM, giving a compact grid-level reference for the recovery pattern.
3. **Reacher is already robust under no-noise-trained PLDM.** The no-noise PLDM drop is only 2.33 pt, and a representative high observation-noise row occurs at $\sigma_{\max} = 0.03$ (81.67%). This makes Reacher a low-cliff second-family reference for interpreting noise-training gains.
4. **Cube (structured 3D manipulation) remains weakly noise-responsive.** PLDM Cube unperturbed success spans 51.33–56.00% and observation-noise 0.08 spans 48.33–54.67% across the full grid. This is the same weak noise responsiveness seen on LeWM.

Method-specific details. PLDM's representative PushT unperturbed row (76.67% at $\sigma_{\max} = 0.03$) is lower than LeWM's corresponding representative row (89.67% at $\sigma_{\max} = 0.03$). This second-family reference is therefore a robustness check: the cliff-and-recovery shape is reproduced, while the exact high-grid location is method-dependent. The LeWM PushT split between unperturbed and noisy representative rows is a LeWM-family signature within this fixed protocol.

PLDM full-sweep ACPC basin replication. We also run the Gaussian-noise basin protocol on all 36 PLDM checkpoints. Table 25 reports the compact baseline-vs-representative observation-noise summary; the full 4 tasks $\times$ 9 configurations grid is listed in Appendix C.10.

Table 25: PLDM full-sweep Gaussian-noise ACPC basin replication, summarized by the no-noise baseline and each task's representative high-noise PLDM observation-noise 0.08 grid row. The protocol matches Table 7: same-state Gaussian-noise views of the observation history (std 0.01–0.08), unperturbed goal, and the same recorded action sequence.

Task	checkpoint	obs 0.08	drop	R_E	R_F
TwoRoom	base	67.00	29.67	3.287	0.611
TwoRoom	noise 0.06	98.00	0.00	0.235	0.034
PushT	base	18.33	57.00	0.874	0.846
PushT	noise 0.03	73.00	3.67	0.153	0.091
Reacher	base	80.33	2.33	0.237	0.060
Reacher	noise 0.03	81.67	1.33	0.135	0.039
Cube	base	48.33	4.33	0.831	0.453
Cube	noise 0.04	54.67	−0.67	0.146	0.054

Reading. The PLDM basin replication is directionally consistent with the LeWM ACPC-basin reading: the reported representative observation-noise 0.08 checkpoint reduces R_F on all four tasks, sharply on TwoRoom, PushT, and Cube. Reacher is the boundary case: PLDM is already near-robust at the no-noise baseline, and both the behavioral drop and the basin change are small. The full 36-row artifact reduces the concern that the paired basin pattern is purely a LeWM artifact, but it does not upgrade the claim to a general mechanism because Table 26 shows different internal diagnostic movement.

PLDM full-diagnostic mechanism check. We also run the same five diagnostic layers on the PLDM sweep. The compact PLDM base-vs-representative view in Table 26 matches the guard reading in Table 10: both families preserve rank, transition resolution, and the inverse-dynamics probe to first order at representative recovered rows. The broader full-diagnostic artifacts also show strong multi-step rollout-drift reductions, but those rollout-drift measurements are mechanism-localization readouts rather than main-text collapse guards. The clearest method-specific difference is TwoRoom: LeWM's fixed-0.08 checkpoint compresses effective rank (47.6 → 37.7), whereas PLDM's representative row leaves rank essentially unchanged (74.8 → 72.7). This separates two claims: the task-level fragility/recovery signature replicates across method families, while the representative diagnostic movement remains architecture-dependent.

Table 26: PLDM full-diagnostic check, no-noise baseline → representative noise-trained checkpoint. Representatives are high-noise PLDM observation-noise 0.08 grid rows (TwoRoom $\sigma_{\max} = 0.06$, PushT $\sigma_{\max} = 0.03$, Reacher $\sigma_{\max} = 0.03$, Cube $\sigma_{\max} = 0.04$).

Task	rep. $\sigma_{\max}$	eff. rank	trans. L2	ID probe R^2	8-step rollout L2
TwoRoom	0.06	74.78 → 72.70	0.8188 → 0.8254	0.3233 → 0.3201	20.36 → 1.02
PushT	0.03	130.13 → 131.90	0.4710 → 0.4778	0.7570 → 0.7479	20.25 → 2.82
Reacher	0.03	117.52 → 108.14	0.5053 → 0.4824	0.2150 → 0.1844	1.45 → 0.94
Cube	0.04	134.80 → 124.49	0.6395 → 0.6313	0.5428 → 0.5576	18.98 → 4.02

What this strengthens, and what remains bounded. PLDM reduces the concern that the four task signatures are tied to one LeWM checkpoint family. The full PLDM basin artifact, summarized in Table 25, further supports the paired same-state perturbation reading at the baseline-vs-representative high-noise grid rows. The PLDM sweep also reinforces the bounded-plateau reading: the exact high-grid location is method- and task-dependent, so the diagnostic tables should localize representative checkpoints rather than rank points inside the sweep.

Mechanism boundary. Table 26 is the reason we do not promote any single internal mechanism to a general explanation. On both families the most consistent representative-checkpoint movement is a strong reduction in multi-step rollout drift, but the detailed movement is method-specific: LeWM compresses TwoRoom's effective rank where PLDM does not. The shared conclusion is therefore about *control-time visual fragility and noise-training recovery*; the internal diagnostic route is method-specific.

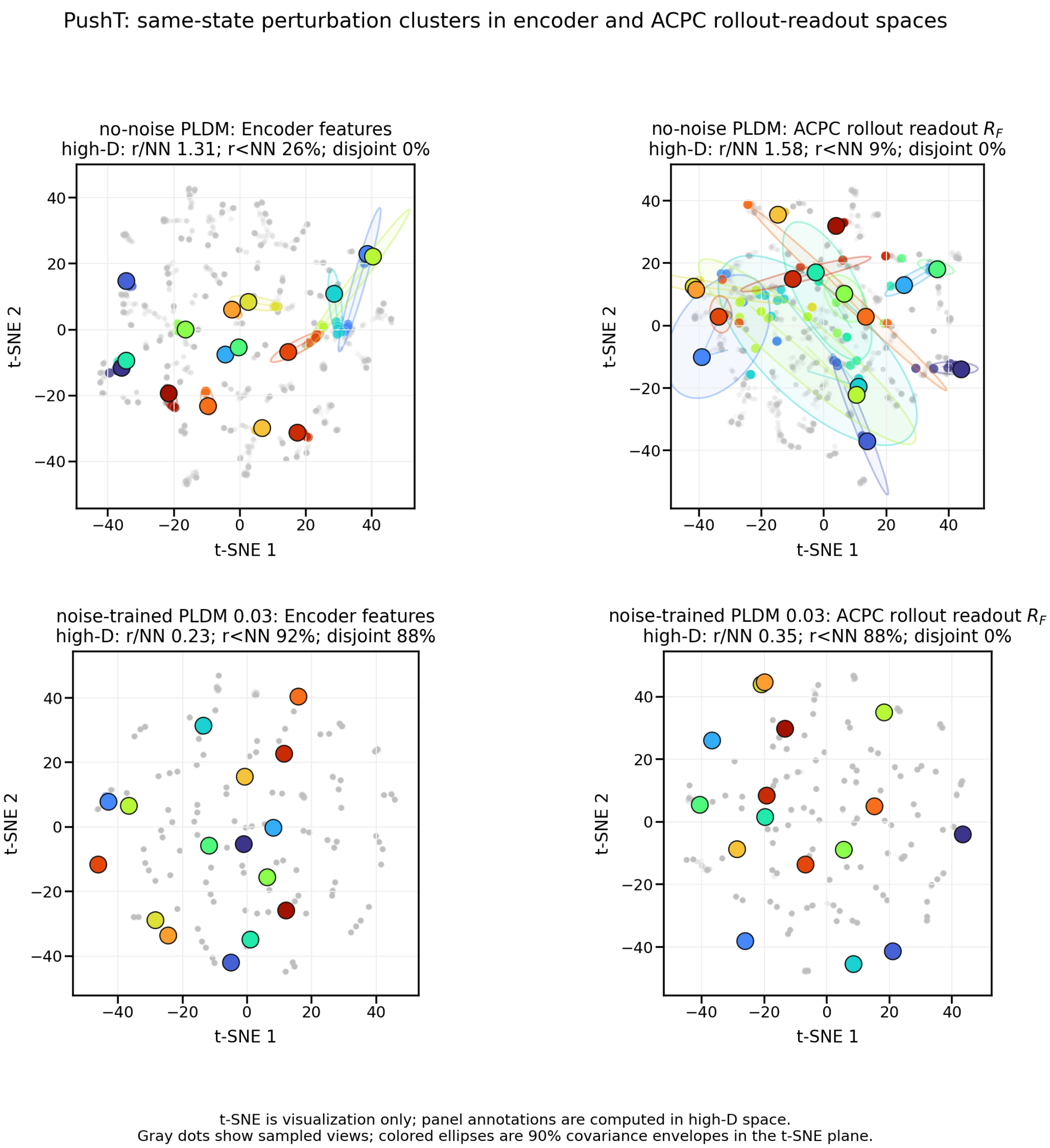


Figure 5: Qualitative PLDM PushT same-state perturbation clusters for Table 25. Top: no-noise-trained PLDM; bottom: noise-trained PLDM at $\sigma_{\max} = 0.03$. Small panel summaries report original-space local statistics; t-SNE coordinates and envelopes are qualitative visualization only and not comparable with LeWM.

J Cross-perturbation sanity check: no-noise-trained blur evaluation

This appendix addresses the narrow concern that the observed visual fragility is an artefact of Gaussian noise only. We evaluate the LeWM and PLDM baselines trained without input-noise augmentation under Gaussian blur with kernel sizes $3, 7, 11, 15$, applied to pixels, goal images, or both. This is an **eval-only** stress test: no blur-trained checkpoint is included, and the blur data are not mixed into the Gaussian-noise sweep. The released source is listed in Appendix C.10.

Table 27: Observation+goal blur stress test on baselines trained without input-noise augmentation. Success rate mean ± population std over 3 evaluation seeds × 100 trajectories. Kernel sizes 11 and 15 are severe low-pass endpoints on 224×224 inputs, so they should not be read as mild perturbations.

Method	Task	unpert.	$k = 3$	$k = 7$	$k = 11$	$k = 15$	worst drop
LeWM	TwoRoom	94.00 ± 3.56	39.33 ± 2.05	32.67 ± 3.09	30.33 ± 2.87	30.67 ± 0.47	63.67
LeWM	PushT	86.33 ± 2.36	81.67 ± 4.50	74.00 ± 1.41	65.33 ± 2.49	58.67 ± 6.65	27.67
LeWM	Reacher	58.67 ± 1.25	57.00 ± 6.38	45.67 ± 2.05	36.00 ± 4.90	20.33 ± 2.49	38.33
LeWM	Cube	66.67 ± 2.62	65.00 ± 1.63	62.33 ± 2.62	57.33 ± 2.87	54.00 ± 2.16	12.67
PLDM	TwoRoom	96.67 ± 1.70	38.33 ± 0.47	30.33 ± 2.87	28.33 ± 1.25	28.33 ± 1.70	68.33
PLDM	PushT	75.33 ± 3.68	74.00 ± 2.94	72.00 ± 3.56	67.67 ± 4.03	62.33 ± 2.05	13.00
PLDM	Reacher	82.67 ± 1.70	86.00 ± 2.16	80.00 ± 3.56	72.00 ± 2.16	68.00 ± 4.08	14.67
PLDM	Cube	52.67 ± 3.30	53.00 ± 5.35	53.00 ± 2.83	50.67 ± 3.68	52.33 ± 5.44	2.00

Reading. Blur is a different failure mode from Gaussian pixel noise. The clear collapse is TwoRoom: it falls below 40% already at $k = 3$ on both LeWM and PLDM and then saturates around 28–33% for $k = 7$–15. This reverses the Gaussian-noise ordering, where PushT is the most fragile task. The other tasks are more stable under blur: PLDM PushT, PLDM Reacher, and both Cube baselines lose at most about 15 pt at the strongest endpoint, while LeWM PushT degrades gradually and LeWM Reacher drops sharply only at $k = 15$. We therefore use blur only as an eval-only cross-perturbation sanity check: visual fragility is not unique to Gaussian noise, but the task ordering and recovery profile are perturbation-specific. Because this appendix does not train blur-robust checkpoints or test Gaussian-trained checkpoints under blur, it does not establish cross-perturbation transfer.

K Unseen-stressor score and matched diagnostic check

After the Gaussian development grid and training-seed replication were frozen, we ran strongest-severity unseen-stressor score checks on all three LeWM training seeds (3072/3073/3074), comparing no-noise training to $\sigma_{\max} = 0.08$ under two unseen visual stressors: Gaussian blur with kernel size 15 and resize/down-upsample with factor 0.25. A compact four-row summary of this appendix appears in Table 12; the appendix gives the full score aggregate and matched diagnostic slice. This appendix does not broaden the main Gaussian-noise claim into a cross-perturbation result. It asks whether the Gaussian-trained endpoint transfers to non-Gaussian stressors, and whether the paired diagnostics move in the same direction when the behavioral effect is clear. The score aggregate and selected diagnostics subset both include all three training seeds, with TwoRoom/Reacher blur as clear positive-transfer endpoints and PushT/Cube resize as small-effect endpoints. The full review file is listed in Appendix C.10.

Table 28: Three-training-seed unseen-stressor score check. Values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from the strongest-severity unseen runs (`num_eval=300`). Stress Δ is $\sigma_{\max} = 0.08$ stress success minus no-noise stress success; drop improvement is the reduction in clean-to-stress drop.

Task	unseen stress	baseline stress	$\sigma_{\max} = 0.08$ stress	stress Δ	drop impr.	reading
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$+43.11 \pm 8.90$	$+40.89 \pm 8.23$	clear score gain
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$+49.22 \pm 3.29$	$+30.22 \pm 4.16$	clear score gain
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$+2.89 \pm 17.98$	-3.78 ± 15.56	no clear gain
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	-0.89 ± 1.40	$+2.78 \pm 3.40$	no clear gain

Reading. TwoRoom and Reacher show the desired agreement: the unseen stress scores improve strongly and the displayed paired diagnostics move in the better direction. PushT and Cube should not be read as negative transfer cases: their resize score deltas are small at the scale of the evaluation variance, so the behavioral conclusion is no clear effect. The diagnostics act as a specificity check for the same cautious interpretation. PushT shows improved ACPC-H/transition and PCC but almost unchanged CRA and no clear stress-score gain; Cube shows small and inconsistent ACPC/PCC/CRA movement. We therefore treat this as bounded appendix evidence for diagnostic localization on selected clear-effect unseen stressors, plus no coordinated score/diagnostic signal on the resize endpoints; it is not evidence that ACPC is a universal cross-perturbation robustness predictor.

L Phase-0 paired ACPC diagnostics

L.1 Exploratory readout definitions

The main text fixes ACPC to the 8-step identity-readout rollout-latent basin diagnostic and reports a compact LeWM subset of the shared-candidate readouts in Section 5.5. The full readouts below remain exploratory and are not used for model selection.

Encoder perturbation difference (EPD).

EPD $= d(E(o), E(\tilde{o}))$. This reports whether a perturbation enters the latent, not whether it matters for control.

One-step consistency (ACPC-1).

The one-step paired score is

$$\mathrm{ACPC}_1 = d\big(F(E(o), a), F(E(\tilde{o}), a)\big).$$

For cross-checkpoint comparisons we optionally divide by the natural transition magnitude $d(z_t, z_{t+1}) + \epsilon$.

Multi-step consistency (ACPC-H).

Equation (3) with $H \in \{4, 8\}$ measures disagreement between unperturbed and perturbed rollouts under the same action sequence.

Predictive cost consistency (PCC).

$\mathrm{PCC} = \big|J(F^H(E(o), \mathbf{a}), g) - J(F^H(E(\tilde{o}), \mathbf{a}), g)\big|$ for a cost/goal readout J and goal g. PCC tests whether predictive mismatch changes the planning cost.

Candidate-ranking agreement (CRA).

For a shared candidate set $\mathcal{A} = \{\mathbf{a}^1, \ldots, \mathbf{a}^K\}$, CRA is the Spearman/Kendall agreement of the unperturbed and perturbed cost vectors.

Margin-conditioned action flip (MAF).

$\mathrm{MAF} = \mathbf{1}\big[u^*_{\mathrm{orig}} \neq u^*_{\mathrm{noise}},\ C_{(2)} - C_{(1)} > \delta\big]$: a flip of the selected action counts only when the unperturbed top-1/top-2 cost margin exceeds δ.

Action-relevant discriminability margin (ADM).

Over action/transition-distinct pairs P_{disc}, $\mathrm{ADM} = \mathrm{median}_{(i,j)\in P_{\mathrm{disc}}}\, d(\Psi_i, \Psi_j)$. In this paper ADM is a proxy check rather than a direct proof of the margin condition in Equation (6).

Selective predictive robustness ratio (SPRR).

A summary of action-distinct discriminability relative to original/perturbed predictive inconsistency,

$$\mathrm{SPRR} = \frac{\mathrm{median}_{(i,j)\in P_{\mathrm{disc}}}\, d(\Psi_i, \Psi_j)}{\mathrm{median}_{P_{\mathrm{pert}}}\, \mathrm{ACPC}_H + \epsilon}.$$

L.2 Phase-0 table

This appendix records the full-coverage clean-goal run of these paired action-conditioned diagnostics. The released artifact covers 72 checkpoint rows: LeWM and PLDM, four tasks, and nine training-noise levels per task. All rows completed successfully. Each row uses 100 sampled sequences, Gaussian observation-history noise at std 0.08 with the goal image kept clean, rollout horizon $H = 8$, and a shared candidate-action set of one dataset future plus 64 random futures. This matches the paper's primary observation-noise endpoint rather than the auxiliary observation+goal stress condition. PCC, CRA, and MAF are computed from the same original/noisy candidate-cost vectors. ADM and SPRR use an action-distance latent proxy, not an oracle state/keypoint margin.

Table 29: Three-seed unseen-stressor matched diagnostic slice with selected Phase-0 paired diagnostics. Values are averages over training seeds 3072/3073/3074. Stress delta is $\sigma_{\max} = 0.08$ minus the no-noise baseline at the strongest unseen stress. Drop improvement is the reduction in clean-to-stress drop. Negative Δ is better for ACPC-H/transition and PCC; positive Δ is better for CRA. Small PushT/Cube score deltas are treated as no clear behavioral effect at this evaluation scale.

Task	stress	stress Δ	drop impr.	Δ ACPC	Δ PCC	Δ CRA	reading
TwoRoom	blur $k = 15$	+43.11	+40.89	−0.691	−44.7	+0.616	clear signal
Reacher	blur $k = 15$	+49.22	+30.22	−1.895	−48.9	+0.587	clear signal
PushT	resize 0.25	+2.89	−3.78	−0.267	−6.7	+0.010	no clear effect
Cube	resize 0.25	−0.89	+2.78	+0.052	−0.6	−0.030	no clear effect

Table 30: Full blur/resize unseen-stressor diagnostic audit. Rows are the 24 task–family–training-seed endpoints: four tasks, blur and resize strongest stressors, and training seeds 3072/3073/3074. Correlations compare diagnostic deltas from $\sigma_{\max} = 0.0 \to 0.08$ against closed-loop strongest-stress score deltas and drop improvements. This table reduces selected-slice risk but remains a bounded two-family scope check.

Metric	ρ vs stress Δ	r vs stress Δ	ρ vs drop impr.	r vs drop impr.	n
ACPC-H/trans. Δ	0.92	0.85	0.77	0.67	24
PCC Δ	0.94	0.90	0.84	0.90	24
CRA Δ	0.94	0.89	0.87	0.91	24
MAF Δ	0.90	0.91	0.77	0.77	24
Composite signed-rank	0.94	0.94	0.82	0.84	24

Table 31: Phase-0 paired ACPC diagnostics, no-noise baseline $\rightarrow$ representative high observation-noise 0.08 checkpoint with an unperturbed goal. Lower is better for ACPC-H/transition, PCC, and MAF; higher is better for CRA and top-8 overlap. Values are exploratory diagnostics.

Task	Method	robust $\sigma_{\max}$	obs-noise success	ACPC-H/trans.	PCC	CRA	top-8	MAF
TwoRoom	LeWM	0.08	65.67 $\rightarrow$ 97.67	1.178 $\rightarrow$ 0.041	42.0 $\rightarrow$ 1.0	0.364 $\rightarrow$ 0.999	0.330 $\rightarrow$ 0.971	0.81 $\rightarrow$ 0.04
TwoRoom	PLDM	0.06	67.00 $\rightarrow$ 98.00	1.100 $\rightarrow$ 0.049	33.6 $\rightarrow$ 0.7	0.416 $\rightarrow$ 0.998	0.335 $\rightarrow$ 0.959	0.79 $\rightarrow$ 0.06
PushT	LeWM	0.06	4.33 $\rightarrow$ 87.67	2.625 $\rightarrow$ 0.146	56.6 $\rightarrow$ 2.6	0.412 $\rightarrow$ 0.997	0.214 $\rightarrow$ 0.931	0.98 $\rightarrow$ 0.08
PushT	PLDM	0.03	18.33 $\rightarrow$ 73.00	1.495 $\rightarrow$ 0.140	23.2 $\rightarrow$ 2.7	0.593 $\rightarrow$ 0.987	0.444 $\rightarrow$ 0.909	0.69 $\rightarrow$ 0.06
Reacher	LeWM	0.02	18.33 $\rightarrow$ 83.67	2.092 $\rightarrow$ 0.029	72.2 $\rightarrow$ 0.5	0.278 $\rightarrow$ 1.000	0.360 $\rightarrow$ 0.989	0.91 $\rightarrow$ 0.01
Reacher	PLDM	0.03	80.33 $\rightarrow$ 81.67	0.122 $\rightarrow$ 0.074	1.9 $\rightarrow$ 1.3	0.994 $\rightarrow$ 0.997	0.955 $\rightarrow$ 0.974	0.13 $\rightarrow$ 0.06
Cube	LeWM	0.07	47.00 $\rightarrow$ 67.33	1.979 $\rightarrow$ 0.058	63.7 $\rightarrow$ 1.3	0.274 $\rightarrow$ 0.999	0.256 $\rightarrow$ 0.970	0.89 $\rightarrow$ 0.00
Cube	PLDM	0.04	48.33 $\rightarrow$ 54.67	1.175 $\rightarrow$ 0.170	27.1 $\rightarrow$ 3.9	0.739 $\rightarrow$ 0.990	0.536 $\rightarrow$ 0.916	0.74 $\rightarrow$ 0.08

Reading. The Phase-0 pass converts the ACPC definitions from a prospective protocol into a release artifact that behaves sensibly on the existing sweep under the same clean-goal stressor used for the primary closed-loop endpoint. The largest Gaussian-noise cliffs (TwoRoom and PushT on both methods, Reacher/Cube on LeWM) coincide with large baseline ACPC-H, PCC, ranking disruption, and action flips, and the representative high observation-noise checkpoints reduce these quantities sharply. PLDM Reacher is the important boundary case: it is already robust at the no-noise baseline, and the paired diagnostics are correspondingly already close to the recovered checkpoint. We therefore treat Table 31 as a face-validity and local-diagnostic check, not as evidence that ACPC-H, PCC, CRA, MAF, ADM, or SPRR can by themselves predict robustness.

M Selector-baseline audit for fixed-rule plateau localization

This audit compares the seed-3072-fixed aggregate ACPC/PCC/CRA/MAF rule against simple selector baselines over the same three-training-seed Gaussian grid. All rows use nonzero training-noise candidates only. The random baseline is the exact expected regret over the eight nonzero grid rows, not a Monte Carlo sample. The released audit artifact is `assets/paper1_data/selector_baseline_audit_20260704.json`.

Table 32: Selector-baseline audit for no-retraining plateau localization. Regret is closed-loop best observation-noise 0.08 success minus selected success. The aggregate rule is comparable to fixed $\sigma_{\max} = 0.08$ and single-metric selectors on the broad Gaussian plateaus; the audit is therefore plateau context rather than evidence that the aggregate rule dominates all obvious baselines.

Selector	split	within 5pp	regret mean ± std	reading
Aggregate ACPC/PCC/CRA/MAF	all	10/12	2.25 ± 2.51	plateau localization
Fixed $\sigma_{\max} = 0.08$	all	10/12	2.14 ± 2.18	competitive simple endpoint
ACPC only	all	9/12	3.39 ± 4.21	weaker on dev seed
PCC only	all	10/12	2.25 ± 2.51	comparable to aggregate
CRA only	all	10/12	2.19 ± 2.56	comparable to aggregate
MAF only	all	10/12	1.89 ± 2.80	comparable or slightly lower regret
Random nonzero std	all	–	7.02 ± 4.01	exact expected regret over eight rows
Aggregate ACPC/PCC/CRA/MAF	held-out 3073/3074	7/8	2.21 ± 1.83	held-out plateau audit
Fixed $\sigma_{\max} = 0.08$	held-out 3073/3074	7/8	2.08 ± 1.93	comparable endpoint baseline
MAF only	held-out 3073/3074	7/8	1.62 ± 1.66	best simple baseline in this audit
Random nonzero std	held-out 3073/3074	–	7.23 ± 4.37	plateau context lower bound check

Reading. The aggregate rule should not be read as a universally superior selector. It is useful because it turns the paired diagnostics into a fixed no-retraining audit that stays near the closed-loop plateau and exposes failure cases; fixed high-noise and MAF-only selectors are equally strong on this grid. Reporting this audit prevents attributing broad Gaussian plateaus to the diagnostic rule alone.

N Target-view ablation completion check

This appendix records the completed LeWM target-view ablation referenced in Section 6. The runs use the origin-future target configuration: the prediction context is encoded from the configured Gaussian-perturbed history, while the prediction target remains the original future latent. This is the perturbed-history → original-future branch, contrasted with the default full-sequence perturbed-target branch used by the main LeWM+noise sweep. The released summary artifact is listed in Appendix C.10.

We verified the four-task run set after completion: eight target-noise checkpoints per task and epoch-10 summaries for unperturbed, observation-noise 0.03, observation-noise 0.05, and observation-noise 0.08 evaluation. The canonical table below uses seeds 42/43/44 and 100 trajectories per seed.

Table 33: LeWM perturbed-history → original-future target-view ablation. Values are success rate mean ± population std over the eight target-noise checkpoints; each checkpoint value is the three-seed mean from `eval_summary.csv`.

Task	unperturbed	obs 0.03	obs 0.05	obs 0.08
TwoRoom	93.33 ± 1.72	88.67 ± 6.04	77.71 ± 9.12	61.75 ± 7.65
PushT	82.92 ± 4.00	57.71 ± 8.96	20.21 ± 11.68	6.75 ± 5.00
Reacher	59.92 ± 1.80	40.38 ± 2.78	32.00 ± 4.57	19.62 ± 3.97
Cube	65.00 ± 2.01	61.42 ± 2.06	44.29 ± 1.32	39.62 ± 1.22

Table 34: Closed-loop target-view comparison at observation-noise 0.08. Values are success rate mean ± population std over the eight training-noise checkpoints; each checkpoint value is the canonical three-seed mean.

Task	full-sequence perturbed target	perturbed-history → original future	full-seq advantage
TwoRoom	94.71 ± 1.90	61.75 ± 7.65	+32.96
PushT	72.83 ± 12.84	6.75 ± 5.00	+66.08
Reacher	76.17 ± 10.89	19.62 ± 3.97	+56.54
Cube	59.83 ± 5.55	39.62 ± 1.22	+20.21

Reading. The ablation acts as a negative scope check and supports retaining the full-sequence empirical branch. Simply changing the prediction target to the original future does not reproduce the recovery from input-side noise augmentation in Table 15: PushT and Reacher remain close to their no-noise robustness cliffs, TwoRoom degrades under stronger eval noise despite high unperturbed success, and Cube does not improve over the main sweep.

A matched PushT 0to008 check shows the contrast: rerunning both branches' $\sigma_{\max} = 0.08$ checkpoints under the same fixed evaluation code and canonical seeds gives observation-noise 0.08 three-seed means of 86.33% for the full-sequence checkpoint versus 8.67% for the origin-target checkpoint (the canonical sweep value for the same full-sequence checkpoint in Table 15 is 89.00%; the difference is normal evaluation variance). We interpret this as evidence consistent with a closed-loop train/eval consistency explanation: full-sequence noise augmentation better matches the planner's latent autoregressive feedback, while one-step origin-target denoising does not. The result narrows the mechanism: target view alone is not the missing ingredient, and an explicit action-conditioned predictive-consistency objective is the method-paper path suggested by the diagnostics.